

ROLE OF INFORMATION TECHNOLOGY IN INDIAN BANK

CHAPTER -1

INTRODUCTION

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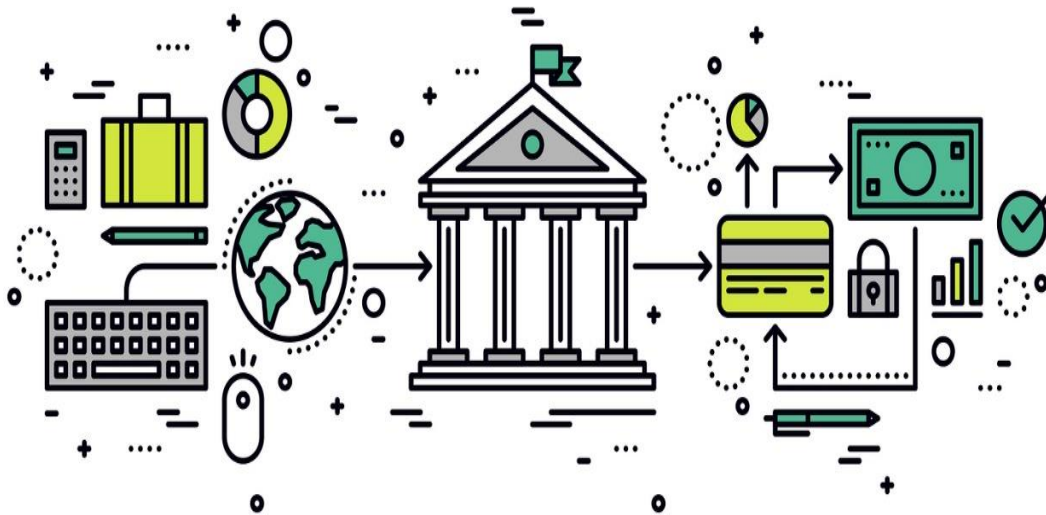
The term “Information technology” refers to the use of sophisticated information and communication technologies together with computer science to enable banks to offer better services to its customers in a secure, reliable, and affordable manner, and sustain competitive advantage over other banks. Indian Banking technology also subsumes the activity of using advanced computer algorithms in unravelling the patterns of customer behavior by sifting through customer details such as demographic, psychographic, and transactional data.

The banks in India are using Information Technology (IT) not only to improve their own internal processes but also to increase facilities and services to their customers. Indian Bank today have become synonymous with technology and have leveraged IT in all areas of governance, operations and control. Effectively use of Technology has facilitated accurate and timely management of the increased volume of banks that comes with a larger customer base.

Indian banking sector is the most dominant sector of the financial system in India. Significant progress has been made with respect to the banking sector in the post liberalization period. The financial health of the commercial banks has improved manifolds with respect to capital adequacy, profitability, and asset quality and risk management. Further, deregulation has opened new opportunities for banks to increase revenue by diversifying into investment banking, insurance, credit cards, depository services, mortgage, securitization, etc. Liberalization has created a more competitive environment in the banking sector.

During the recent years, the pace and quality of banking was changed by the technological advancements made in this area. Computerization as well as the

adoption of core banking solution was one of the major steps in improving the efficiency of banking services. The process of computerization of the banking sector continued.



Objectives can be attributed to it

1. The use of appropriate hardware for conducting business and servicing the customers through various delivery channels and payment systems and the associated software constitutes one dimension of banking technology. The use of computer networks, security algorithms in its transactions, ATM and credit cards, Internet banking, Telebanking, and mobile banking are all covered by this dimension. The advances made in information and communication technologies take care of this dimension.
2. On the other hand, the use of advanced computer science algorithms to solve several interesting marketing-related problems such as customer

segmentation, customer scoring, target marketing, market-basket analysis, cross-sell, up-sell, and customer retention faced by the banks to reap profits and outperform their competitors constitutes the second dimension of banking technology. This dimension covers the implementation of a data warehouse for banks and conducting.

3. Moreover, Indian bank cannot ignore the risks that arise in conducting business with other banks and servicing their customers, otherwise their very existence would be at stake. Thus, the quantification, measurement, mitigation, and management of all the kinds of risks that banks face constitute the third important dimension of banking technology. This dimension covers the process of measuring and managing credit risk, market risk, and operational risk. Thus, in a nutshell, in ‘banking technology’, ‘banking’ refers to the economic, financial, commercial, and management aspects of banking, while ‘technology’ refers to the information and communication technologies, computer science, and risk quantification and measurement aspects.
4. Over a decade Indian banking system witnessed metamorphosis. The main driver of transformation has been the fast adoption of Information, Communication and Technology (ICT) based system in the banks. The huge red ledgers, row of racks of ledger holders, cash scrolls, registers, clearing cheque scrolls, totaling machines, long rolls of paper ribbons often gazing the floor formed part of hardware in the branches.
5. It was also common to see staff hiding behind the tall branch counters, row of signature cabinets standing between the counters and supervisory staff, customers eyeing frantically on movement of ledgers and cheques until their

transactions were done. They are now no more relevant. The banking work space has changed for good. Indian Bank branches are now sporting a smart look with refurbished interior, radiating corporate color, well dressed bank logos, wide glass doors, and plush interiors and well-developed customer lounges etc.

6. The onsite ATMs, teller counters, swipe machines / kiosks have speed up standard transactions of every day need of consumers. With the onset of alternative delivery channels, even the branch timings are not very significant. Phone and mobile banking, smart cards, debit cards, rechargeable electronic purse are also some of the modern-day banking facilities that allow round the clock access. With the profile and aptitude of bank consumers fast changing toward the use of ICT facilities, the popularity of e-channels of banking are set to assume more significance. Banks are fast gearing up to introduce add-on services to attract young generation of customers.
7. The low height counters handled by trained employees wearing inviting look, customers having one to one interface with departments, banking halls buzzing with clicks of mouse, laptops, computers, currency notes zipping through the counting machines form part of modernized attire of bank branches at least in metro cities. Banking and technology go hand-in-hand these days.



Evolution of Banking

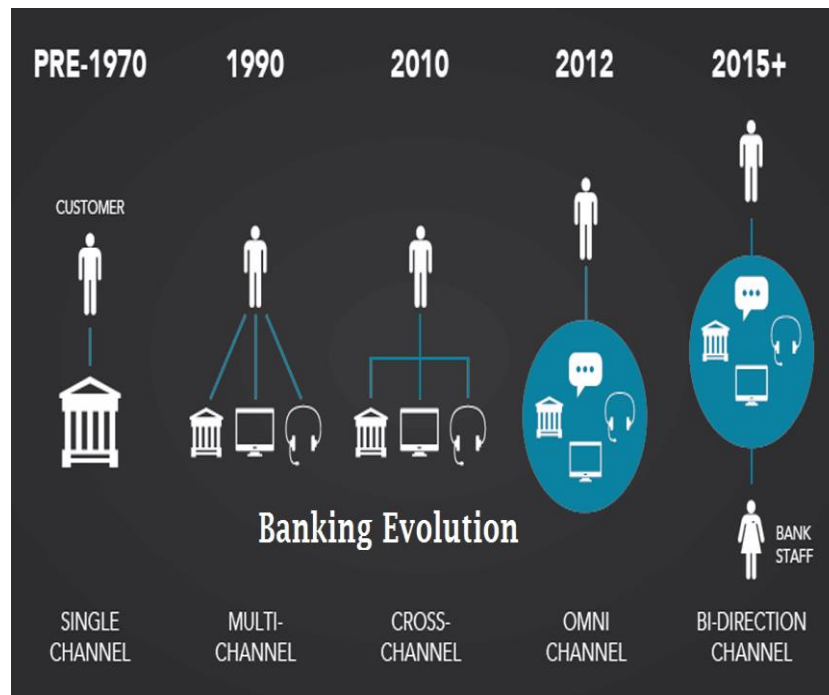
Despite the enormous changes the banking industry has undergone through during the past 20 years let alone since 1943 one factor has remained the same: the fundamental nature of the need customers have for banking services. However, the framework and paradigm within which these services are delivered has changed out of recognition. It is clear that people's needs have not changed, and neither has the basic nature of banking services people require. But the way banks meet those needs is completely different today. They are simply striving to provide a service at a profit. Banking had to adjust to the changing needs of societies, where people not only regard bank account as a right rather than a privilege, but also are aware that their business is valuable to the bank, and if the bank does not look after them, they can take their business elsewhere (Engler & Eslinger, 2000).

Indeed, technological and regulatory changes have influenced the banking industry during the past 20 years so much so that they are the most important

changes to have occurred in the banking industry, apart from the ones directly caused by the changing nature of the society itself. In this book, technology is used interchangeably with information and communication technologies together with computer science. The relationship between banking and technology is such that nowadays it is almost impossible to think of the former without the latter. Technology is as much part of the banking industry today as a ship's engine is part of the ship. Thus, like a engine, technology drives the whole thing forward (Engler &Eslinger, 2000). Technology in banking ceased being simply a convenient tool for automating processes. Today banks use technology as a revolutionary means of delivering services to customers by designing new delivery channels and payment systems. For example, in the case of ATMs, people realized that it was a wrong approach to provide the service as an additional convenience for privileged and wealthy customers.

It should be offered to the people who find it difficult to visit the bank branch. Further, the cost of delivering the services through these channels is also less. Banks then went on to create collaborative ATM networks to cut the capital costs of establishing ATM networks, to offer services to customers at convenient locations under a unified banner (Engler &Eslinger, 2000). People interact with banks to obtain access to money and payment systems they need. Banks, in fact, offer only what might be termed as a secondary level of utility to customers, meaning that customers use the money access that banks provide as a means of buying the things they really want from retailers who offer them a primary level of utility. Customers, therefore, naturally want to get the interaction with their bank over as quickly as possible and then get on with doing something they really want to do or with buying something they really want to buy. That explains why new types of delivery channels that allow rapid, convenient, accurate delivery of banking services to customers are so popular. Nowadays, customers enjoy the fact that their banking chores are done quickly and easily (Engler &Eslinger, 2000). The kind of

enormous and far-reaching developments discussed above have taken place along with the blurring of demarcations between different types of banking and financial industry Activities



1. Governments have implemented philosophies and policies based on an increase in competition in order to maximize efficiency. This has resulted in the creation of large new financial institutions that operate simultaneously in several financial sectors such as retail, wholesale, insurance, and asset management.
2. New technology creates an infrastructure allowing a player to carry out a wide range of banking and financial services, again simultaneously.
3. Indian Banks had to respond to the increased prosperity of their customers and to customers 'desire to get the best deal possible. This has encouraged banks to extend their activities into other areas.

4. Indian Banks had to develop products and extend their services to accommodate the fact that their customers are now far more mobile. Therefore, demarcations are breaking down.

5. Indian Banks have every motivation to move into new sectors of activity in order to try to deal with the problem that, if they only offer banking services, they are condemned banks realized the convenience of ATMs, new services started to be added.

CHAPTER 2

COMPANY PROFILE

INDIAN Bank



Indian Bank is an Indian public sector bank, established in 1907 and headquartered in Chennai. It serves over 100 million customers with 40,187 employees, 5,880 branches with 4,937 ATMs and Cash deposit machines. Total business of the bank has touched ₹1,221,773 crore (US\$140 billion) as of 31 March 2024.

The bank's information systems and security processes are certified to meet ISO27001:2013 standard. It has overseas branches in Colombo and Singapore including foreign currency banking units in Colombo and Jaffna. It has 227 overseas correspondent banks in 75 countries. Since 1969, the Government of India has owned the bank. As per the announcement made by the Indian Finance Minister Nirmala Sitharaman on 30 August 2019, Allahabad Bank merged on 1 April 2020, making Indian Bank the seventh largest bank in India.

HISTORY

In the last quarter of 1906, Madras (now Chennai) was hit by the worst financial crisis the city was ever to suffer. The three best-known British commercial names in 19th-century Madras, one crashed; a second had to be resurrected by a distress sale; and the third had to be bailed out by a benevolent benefactor. Arbuthnot & Co, which failed, was considered the soundest of the three. Parry's (now EID Parry), may have been the earliest of them, and Binny and Co.'s founders may have had the oldest associations with Madras, but it was Arbuthnot, established in 1810s, that was the city's strongest commercial organization in the 19th century. A key figure in the bankruptcy case for Arbuthnot's was the Madras lawyer, V. Krishnaswamy Iyer who founded the Indian Bank, which was an offshoot of nationalistic fervour and the Swadeshi movement, when the then British Arbuthnot Bank collapsed and the Indian Bank emerged. Mr V. Krishnaswamy Iyer solicited the support of the Nagarathar Chettiar's authored by Mr. Ramasamy Chettiar, who was Annamalai Chettiar's elder brother. Sri V. Krishnaswamy Iyer and Mr. Ramasamy Chettiar were one of the first directors of Indian Bank. Later on, in 1915, Mr. Annamalai Chettiar was inducted into the board of the Indian Bank. It commenced operations on 15 August 1907, with its head office in Parry's Building, Parry Corner, Madras.

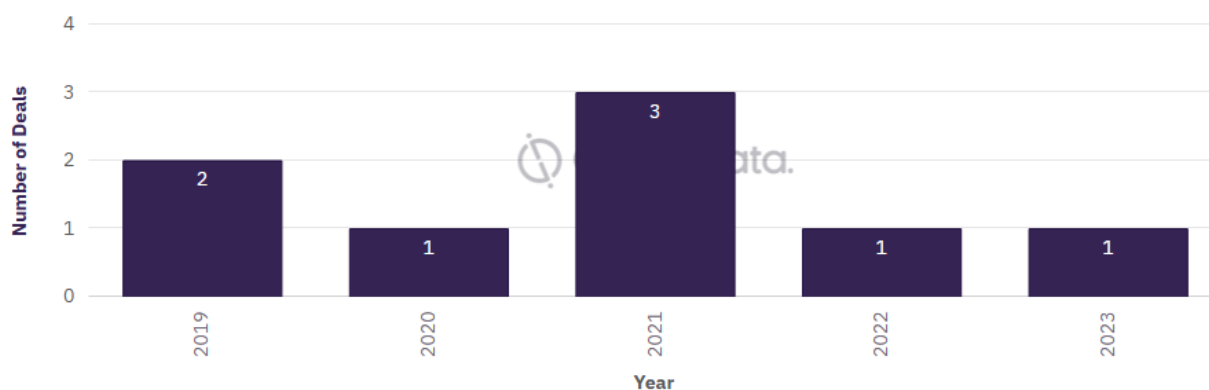
In 1932 IB opened a branch in Colombo. It opened its second branch in Ceylon in 1935 at Jaffna, but closed it in 1939. IB next opened a branch in Rangoon, Burma, in late 1940s. Then in late 1941 IB opened branches in Singapore, Kuala Lumpur, Ipoh, and Penang. The exigencies of war forced IB to close its Singapore and Malayan branches with months. The closing of the Singapore branch resulted in little loss to IB; the loss of the branches in Malaya was much more costly. World War II resulted in further financial problems for IB and it was forced in 1942 to close a number of its branches in India, and also its branch in Colombo.

Competitor Comparison

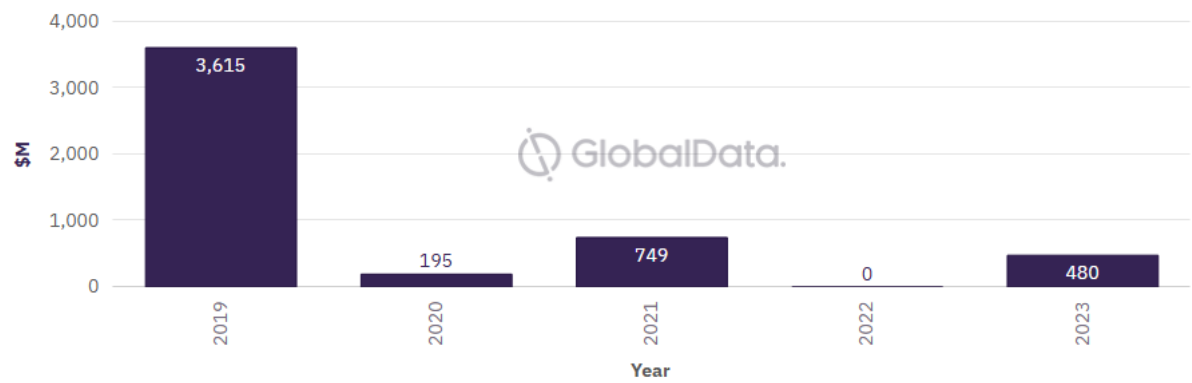
Key Parameters	Indian Bank	Bank of Baroda	Canara Bank	Bank of India	IndusInd Bank Ltd
Headquarters	India	India	India	India	India
City	Chennai	Baroda	Bangalore	Mumbai	Mumbai
State/Province	Tamil Nadu	Gujarat	Karnataka	Maharashtra	Maharashtra
No. of Employees	40,251	75,515	82,638	50,944	45,637
Entity Type	Public	Public	Public	Public	Public

Key Financial Charts

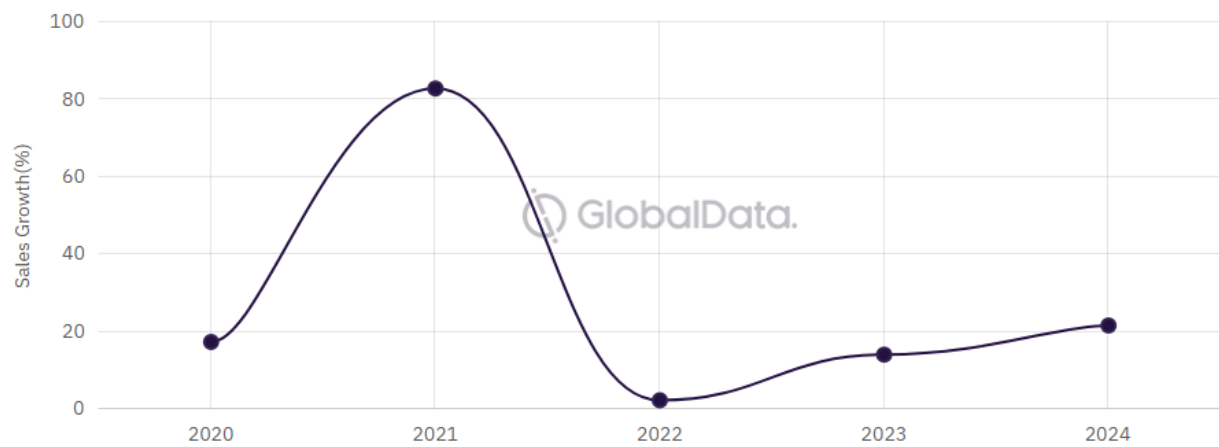
Number of deals



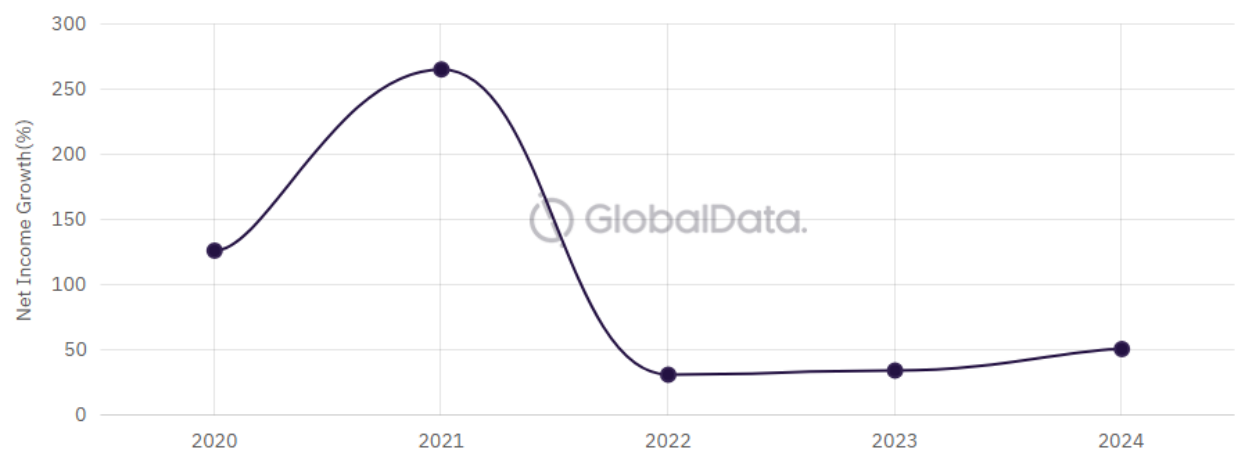
Deal value



Sale growth



Net income



CHAPTER 3

LITERATURE REVIEW

Information Technology (IT) is very powerful in today's world, and financial institutions are the backbone of the Indian economy. Indian Banking Industry today is in the midst of an IT revolution. Nearly, all the nationalized banks in India are going for information technology-based solutions. The application of IT in Banks has reduced the scope of traditional or conventional banking with manual operations. Nowadays banks have moved from disbursed to a centralized environment, which shows the impact of IT on banks. Banks are using new tools and techniques to find out their customers need and offer them tailor made products and services. The impact of automation in banking sector is difficult to measure.

A literature review is important due to the following reasons:

- 1) A literature review gives more knowledge about the area in which the research is conducted
- 2) It helps to refine the research topic by determining the research gap.
- 3) It helps to avoid errors of duplication
- 4) It helps to identify the contribution that one's research will make and also provides a justification for the study.
- 5) It will help in understanding how already existing research findings have been presented in that particular area.
- 6) Application of IT in banking
- 7) IT framework for Indian banking
- 8) Technological developments in cooperative banks
- 9) Indian banking sector: challenges and opportunities

The review has been conducted in the following manner:

- 1) First, several literature sources in the area of behavioral finance were identified and studied.
- 2) The topic was then narrowed down upon as there were several discussions on the various factors
- 3) Therefore, I decided to study all such factors and read more articles on this topic which completed the literature review



Technological Development in the Banking Sector

The technological development in the banking sector began with the use of Advanced Ledger Posting Machines (ALPM) in the 1980s and nowadays banks are using core banking solution (CBS) for providing better services to their customers. Over the years several studies have been conducted both at the industry and academic level to examine the impact of IT on banking productivity and profitability.

Prabhakar Rao Ch. (Jan, 2024).

Indian banking in 2010|| IBA Bulletin Special Issues, in this study discussed about the revolutionary changes that witnessed in the financial sector around the world. He stated that net worked branches. ATMs, technology-based payment and settlement system, technology vision of RBI, floating rate of interest have changed the Indian banking sector. He concluded that brick and mortar bank branches will disappear and customers will be able to operation their accounts through electronic devices.

Palani and Yasodha P. (Apr 2023)

The research paper is focused on customer's perceptions on mobile banking offered by Indian Overseas Bank and it also focuses on the various drivers that drive mobile banking consumers. The results of this study showed that gender, education and income of the consumers play an important role in usage of mobile banking. Most of the researches are focused on the acceptance of the mobile banking technology due to which not much research has been conducted on people. The research reveals that if skills can be upgraded among the consumers there will be greater willingness on the part of consumers toward the use of Mobile banking. Some the factors like security trust, gender, education, religion, and price can have minimal effect on consumer mindset towards Mobile banking compared to the other factors.

Achana Sharma (2023)

This paper examines consumer adopting mobile banking as a new electronic payment service .It also focuses on the various factors influencing the adoption of mobile banking in India. When it comes to the research methodology used in the study, data collected has been grouped into two main categories – primary and secondary data. The secondary data have been collected from the newspapers, journals, magazines, internet and also various other research papers..In case of

questionnaires the has been targeted on user and non user of mobile banking which included the Businessmen, servicemen, professionals, students etc. The primary data for the study is extracted from a survey conducted in Ghaziabad in U.P, India. The research had a total of 100 respondents participating in the data collection for understanding the use of Mobile banking. From the data collected it was possible to make projections in the research.

Thakur, Rakhi; Srivastava, Mala. (2023)

The paper studies the factors influencing the adoption intention of mobile commerce. Perceived usefulness, perceived ease of use and social influence are found to be significant dimensions of technology adoption readiness to use mobile commerce while facilitating conditions were not found to be significant. The results of the research study also indicate the perceived credibility risk defined by security risk and privacy risk are significantly associated with behavioural intention in negative relation, which indicates that security and privacy concerns are important in deterring customers from using mobile commerce. This research study developed an integrated model for behavioural intention towards financial innovations. Practical implications of this study is one of the few empirical studies which have investigated the adoption of mobile commerce in India, which is considered one of the fastest growing countries in terms of mobile usage. The study relates to inclusion of both utilitarian and credibility aspect of adoption intention. It gives an empirical basis on which mobile and banking companies can base their mobile payments marketing strategy.

Kumar, Reji G; Rejikumar, G; Ravindran, D Sudharani.

This research paper examines the factors influencing the continuance decisions of the early adopters of m-banking services in Kerala, India. The study used constructs adopted from Technology Acceptance Model along with constructs

of perceived service quality, perceived credibility and perceived risk to empirically establish the influence on satisfaction and continuance usage intentions. The study confirmed that after adoption of the technology, the customer finds satisfaction in the quality parameters of the service. Perceptions about the risks involved in m-banking had adverse impact on service quality and satisfaction.

Ashish Adholiya, Pankaj Dave, Shilpa Adholiya (2022)

This paper investigates the determinants influencing the customer satisfaction for mobile banking users. Customer satisfaction is one of the fundamental marketing constrain in the last three decades. This research is focused to those respondents who are using the mobile banking services by their service provider. For the research 100 respondents are identified. The respondents belong from both private and public sector banks of Udaipur, Rajasthan. The opinions of the respondents were collected using structured questionnaire. Data collected were analyzed using tools like factor analysis, chi-square and correlation analysis. In factor analysis varimax rotation is used and correlation matrix is used for identifying the relationship between the service quality, perceived value, flexibility, technological innovation, brand perception, strategic endorsement and functional performance of mobile banking service with customer satisfaction.

Kalaiarasi, H & Srividya, V. 3 (Jul-Sep 2022)

Mobile banking as a new channel to the existing banking channels provides convenient and cost efficient banking services anytime anywhere. It is observed that, though India has strong potential for mobile banking only 5% of mobile subscribers are registered users of mobile banking. Attracting the new customers may not be easy than retaining the existing mobile banking customers (2009). Hence the current research focuses on the factors influencing actual usage of mobile banking services. The results show that, Indians mobile banking usage is influenced by ease of mobile banking technology, its suitability to the user's

lifestyle and the benefits like mobility and mobile transactions. However customer's perception towards security of mobile transactions and privacy fears demotivates actual usage.

Thomas et al. (2022)

Stated that although technology opens up new dimensions of scope and timing but it creates the possibility for crimes to be committed very quickly. Technology provides benefits for banks but it worsens traditional banking risks. As the amount of products and services offered by technology grows rapidly, consumers are more and more concerned about security and privacy issues. The banking industry has declared information privacy and security to be major obstacles in the development of consumer electronic commerce. Continuous vigilance and revisions will be essential as the scope of technology on banking increases. However, the ease with which capital can potentially be moved between banks and across borders in a technology environment pose a greater sensitivity to economic policy management.

Tenkasi Taluk & Devasena, S Valli, (Jan 2021)

Banking system is the backbone of the economy and Information Technology (IT) in turn has become the backbone of banking activities. Technology, which was playing a supportive role in banking, has come to the forefront with the ever-increasing challenges and requirements. Technology to start with was a business enabler and now has become a business driver. The Banks cannot think of introducing a financial product without IT support. Be it customer service, transactions, remittances, audit, marketing, pricing or any other activity in the Banks, IT plays an important role not to complete the activity with high efficiency but also has the potential to innovate and meet the future requirements. The Banking Sector was early adopter of technology and in that way set an example to the other industries the need to opt for automation for taking full advantage in operational efficiency.

Laukkanen & Tommie (2021).

The aim of the paper is to explore and compare customer value perceptions in internet and mobile banking. The results indicate that customer value perceptions in banking actions differ between internet and mobile channels. The findings suggest that efficiency, convenience and safety are salient in determining the differences in customer value perceptions between internet and mobile banking. By understanding how and what kind of value different service channels provide for customers service providers are better enabled to create actions to enhance internet and mobile banking adoption. The contribution of the paper lies in achieving a more profound understanding on consumer value perceptions to internet and mobile banking.

Goswami, Divakar; Raghavendran, Satish. (2020)

The research is conducted to determine the potential that mobile banking provides for both the banks and the mobile carriers. After the secondary research the report gives an insight into the best-practices based on a critical evaluation of partnership models. Banks and mobile carriers have tested these waters timidly, and many of the resulting offerings were expensive to the banks and mobile carriers and less than enticing to their customers. This report weeds out ineffective partnering models that companies stumble into on their way to developing mobile-banking and identifies the keys to successful partnerships.

Dr. Vinod Kumar Gupta RenuBagoria&NehaBagoria .

This research paper try s to identify and investigate the various factors which influence the customer's decision to use a specific form of mobile banking and specially focus on the evaluation of SMS-based mobile banking in India. The study also plans to connect the gap of research in the acceptance of mobile banking among the customers. The main challenges involved in the adoption of mobile banking are related to the Positive and Negative factors which influence the adoption of SMS-based mobile banking .Second challenge is Focused on the

adoption of mobile banking services by customers and usage of mobile banking in India. Third is related to the different Technologies behind Mobile Banking. The study has its own limitations but the implications and conclusion from the results can provide practical recommendations to the banking areas and banking industries. It can also provide directions for further work

Prerna Sharma Bamoriya (2020)

The study was conducted to identify certain issues relating to banks, mobile handsets and telecom operators, mobile handset operability, security/privacy, standardization of services, customization, Downloading & installing application software and Telecom services quality. For this purpose a descriptive design was adopted to empirically explore the selected issues. Study suggested that from consumers 'perspective mobile handset operability security or privacy and standardization of services are the critical issues. The objective of the research is to study the selected issues in mobile banking form urban customers' perspective and to explore the perceived utility of mobile banking in comparison to retail banking and online banking among the mobile banking users and non-users. The study is aimed to evaluate perceptions and opinions of urban mobile banking users. For this purpose a cross sectional descriptive design was adopted with ad-hoc quota sampling. Sample for the study comprised of 50 mobile banking users and 50 non-users in Indore city, India.

Shastri R.V, (March, 2019)

“Recent trends in Banking Industry| IT emergence, Chartered Financial Analyst, (in this article stated that liberalization policy and intense competition keeps every banker on his toes. Implementation of Information Technology (IT) helps for maintaining proper accounts especially in decision making process. He also stated that facilities like ATM, anywhere banking, Internet and mobile banking

have imported customer service which in turn helps for better customer relations management. He also explained the challenges faced by banks because of IT implementation like employment problem and security concerns. He suggested that the customer delight is the primary goal of all future IT initiatives.

Arora. K.

Highlighted the significance of bank transformation. Technology has a definitive role in facilitating transactions in the banking sector and the impact of technology implementation has resulted in the introduction of new products and services by various banks in India.

Brett

Studied the changing in old money structure into E-Money. Now days the banks are providing different cards (Smart Card, Credit & Debit Cards) to their customers.

O'Leary et al.(2018)

Two issues come to mind when banks talk about security. They are privacy and security, controlling who gets access to the bank's computer system and its programs, and what time to access it. Studies regarding technology on banking examined barriers such as, security, privacy, and trust of Web system (Rotchanakitumnuai and Speece,2003).To be more precise, lack of privacy and security were found to be significant obstacles to the adoption of technology on banking services(Sathye,1999). Challenges on technology is inevitable, therefore care must be taking in since its negative effect can cause the bank billions of money. Breaches of security and disruptions to the system's availability can damage a bank's reputation; this can potentially affect other technology banking services and its usage (Schechter, 2002)

CHAPTER 4

RESEARCH METHODOLOGY

DATA COLLECTION: PRIMARY METHOD

This study attempts to investigate the impact of electronic banking in the Idea. It explores efficiency, profitability and barriers to the development of electronic banking in India. A quantitative methodology and econometrics models will be employed to address the research questions and the hypotheses. This study is based on secondary and primary data. The required data have been collected from various sources i.e. the primary data were obtained through questionnaires and were complemented with oral interviews of experts of IT section and managers of banks involved in the study. Information and data relating to different banking ratios, banking performance, volume and number of e-banking transactions and facilities, trend and progress and different reports and guidelines have been collected from the various annual reports of RBI, Indian bank's association, annual reports of selected banks, reports of bank for international settlement, reports of institute for development and research in banking technology and IRB bulletins.

This includes 10 major commercial banks of India, State Bank of India (SBI), Punjab National Bank (PNB), Yes Bank, Kotak Mahindra Bank, IndusInd Bank, Bank of Baroda, Bank Of India , INDIAN Bank, HDFC Bank and Axis Bank. The annual balance sheet and income statement used were taken from different reports of Reserve Bank of India.

In the literature in the field, there is no consensus regarding the inputs and outputs that have to be used in the analysis of the efficiency of the activity of commercial banks (Berger and Humphrey, 1997). In the studies in the field, five approaches for defining inputs and outputs in the analysis of the efficiency of a bank were developed, namely: the intermediation approach; the production approach; the asset approach; the user cost; the value added approach. The first three approaches

are developed according to the functions that banks do fulfill (Favero and Papi, 2019). The production and the intermediation approaches are the best known ones and the most used in the quantification of bank efficiency (Sealy and Lindley, 2017).

In the production-type approach, banks are considered as deposit and loan producers and it is assumed that banks use inputs such as capital and labour to produce a number of deposits and loans. According to the intermediation approach, banks are considered the intermediaries that transfer the financial resources from surplus agents to the agents with deficit. In this approach it is considered that the bank uses as inputs: deposits, other funds, equity and work, which they transform into outputs such as: loans and financial investments. The opportunity for using each method varies depending on circumstances (Tortosa- Ausina, 2022). The intermediation approach is considered relevant for the banking sector, where the largest share of activity consists of transforming the attracted funds into loans or financial investments (Andrie and Cocris, 2021). In our analysis we will use the following set of inputs and outputs to quantify the efficiency of banks in India:

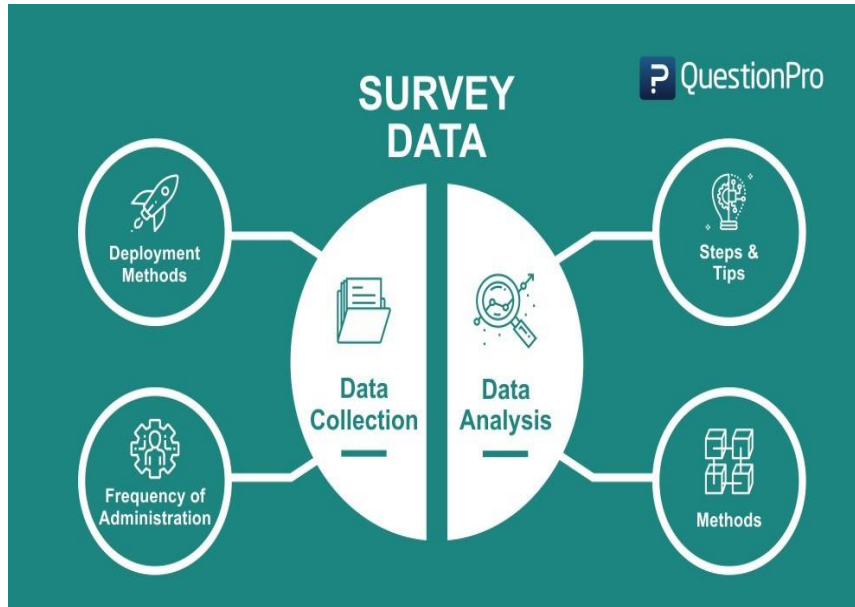
- **Outputs:** Loans and investments
- **Inputs:** Fixed assets, deposits, number of employees, number of branches and number of **ATMs**

This study uses the intermediation approach to define Indian bank inputs and outputs. Under the intermediation approach, banks are treated as financial intermediaries that combine deposits, labour and capital to produce loans and investments. Under the intermediation approach, banks are treated as financial intermediaries that combine deposits, labour and capital to produce loans and investments. In order to measure efficiency of banks we employed DEAP Version 2.1 software.

For identification of obstacles and challenges of development of electronic banking we employed a descriptive and survey research method and also a multistage sampling method is used as sampling method. In this case we found out and accurately described the factors that influence implementation and development of electronic banking.

In the first step of analysing we used bivariate correlation analysis which describes the strength, direction and assessing the significance level of the linear correlation between two variables. These features will help us to test our hypotheses. There are numbers of different statistics available from SPSS in order to test the hypotheses, but depending on the level of our measurements all of which are in interval level we used Pearson Product-Moment Correlation coefficient to test our hypotheses. In order to evaluate research questions independent sample t-test, one way ANOVA test and post-hoc are used to evaluate research questions. In the table provided by Pearson product-moment correlation coefficient there are a number of different aspects of the output that should be considered. Herein, the first thing to consider is assessing the significance level to test the hypotheses.

If the Sig. value is less than 0.05, then with 95% confidence there is a correlation between two variables and consequently Null hypotheses is rejected and the alternative hypothesis is accepted. The value is less than 0.01, then with 99% confidence there is a correlation between variables and again Null hypothesis is rejected. Finally, if the Sig. Value is greater than 0.05, then we conclude that there is no relationship between variables and accordingly the Null hypothesis is accepted. In order to determine the direction of the relationships the negative or positive sign in front of r value will be considered. A negative sign means there is a negative correlation between two variables (i.e. High scores on one variable is associated with low scores on the other) and a positive sign means there is a positive correlation between the two variables (i.e. High scores on one variable is associated with high scores on the other).



SECONDARY METHOD

This study is based on secondary quantitative data and on panel dataset which covering 10 top commercial banks of India over the period of 2015-16 to 2019-20. The banks are State bank of India (SBI), Bank of India (BOI), Bank Of Baroda(BOD) , Punjab national bank (PNB), Yes Bank , INDIAN Bank, HDFC bank, and Axis Bank. The data are from banks tables published by RBI and selected banks.

MODEL SPECIFICATION

The model which we used in this section is based on SCP theory is:

$$ROA_{it} = \alpha_i + \beta_1 IMC_{it} + \beta_2 SIZE_{it} + \beta_3 ATM_{it} + \beta_4 MEMBER_{it} * \\ ATM_{it} + \varepsilon_{it}$$

Where (t) is period of time from 2015 to 2020 and (i) is the number of banks and other variables of study are:

Dependent variable:

1. **ROA:** The return on assets (ROA) percentage shows how profitable a company's assets are in generating revenue. Independent variables:

1. **Indian Bank Index of Market Concentration:** Proponents of banking sector concentration argue that economies of scale drive bank mergers and acquisitions (increasing concentration). Thus, increased concentration goes hand-in-hand with efficiency improvements. Market concentration is one of the dimensions of the banking market. This arguably is the most important structural variable in the equation of profitability. For measurement of concentration in this study we employed Herfindahl–Hirschman Index. The Herfindahl index (also known as Herfindahl–Hirschman Index or HHI) is a measure of the size of firms in relation to the industry and indicator of the amount of competition among them.

$$IMC_{it} = \left[\frac{\text{Total deposit of bank}_i \text{ in year}_t}{\text{Total deposit of all banks in year}_t} \right]^2$$

2. **Size (BSIZE):** Size of bank is another important structural variable which affects profitability of banks. It is believed that big banks due to having more opportunities as compared to small banks are in a better position and their profitability is higher. This variable in this study is defined as:

$$Bsize_{it} = \text{Total deposit of bank}_i \text{ in year}_t$$

3. **Number of ATMs:** Number of ATMs is another independent variable in this study. Due to lack of data for other e-banking services we use only numbers of ATMs as representative of e-banking and period of study will be from 2003-04 to 2010-11.

4. Member to National Financial Switch: Membership to the country's national financial switching is a dummy variable in this model and is used to identify whether membership to financial switching has any impact on profitability of banks. The Institute of Development and Research in Banking Technology (IDRBT) in Hyderabad has been providing the ATM switching service to banks in India through National Financial Switch since 2005.

$$MEMBER_{it} = \begin{cases} 0 & \text{if bank } i \text{ not be member of network in year } t \\ 1 & \text{if bank } i \text{ be member of network in year } t \end{cases}$$

TECHNOLOGICAL DEVELOPMENT IN BANKING

Wave of technology in banking:

The technological development in banking can be traced as follows: -

1960 - Mechanized banking introduced.

1970 - Introduction of computer-based banking industry.

1980 - Introduction of computer-linked communication-based banking.

Advent of computer technology has created a major impact on working of banks. The computerization and subsequent development in history of Indian banks can be traced back to 1966 when Indian Bankers Association (**IBA**) along with exchange banks Association signed first wage settlement with the union, which accounted for the use of IBM or ICT accounting machines for inter-branch reconciliation etc. As per the reports of RBI the first wave in banking technology began with the use of Advanced Ledger Posting Machines (ALPM) in the 1980s. The RBI advised all the banks to go in for huge computerization at the branch level.



There were two options: Automate the front office or the back office. Many banks opted for automating the front office. In the first phase, whereas banks like State Bank of India also concentrated on the back-office automation at the branch level. The Second wave of development was Total Branch Automation (TBA) which came in late 1980s. This automated both

Front-end and Back-end operations within the same branch. TBA comprised of total automation of a particular branch with its own database. In the third wave, the new private sector banks entered into the field of automation. These banks opted for different models of having a single centralized database instead of having multiple databases for all their branches. This was possible due to the availability of good network infrastructure. Earlier, banks were not confident of running the whole operation through single data center.

However, when a couple of private sector banks showed that it could be done efficiently, other banks began to show interest and they also began consolidating their databases into a single database. The banks followed up on this move by choosing suitable application software that would support centralized operations. The fourth wave started with the evolution of the ATM delivery channel. This was the first stage of Empowerment of the customer for his own transactions. The

second stage was the Suvidha experiment in Bangalore. This showed the power of technology and how the reach can be increased amazingly at a great pace. Seeing these, all the banks started revamping their retail delivery channels. Their core focus became increasing the number of customers they can service at a lower cost. The main channels for these were internet banking and mobile banking.

After this, came the alliances for payment through various other gateways. The third important development happening now is the real-time gross settlement system of the RBI. Once this was in place, transactions between banks could be done through the settlement system, online, electronically thereby, ensuring faster collection. The process of computerization had started from back Office application, after that Total Branch Automation and nowadays it is the period of implementation of Core Banking Solutions (CBS).

A key trend in the last couple of years has focused core banking systems. With the implementation of core banking systems across the banks, the usage level of IT for customer management has increased. Core banking systems have enabled banks to launch new products and services targeting specific customer segments after understanding their banking and investment requirements. ATM, internet banking and mobile banking have Improved customer convenience by providing anywhere any time banking services. The utility bill presenting and payment has help customers to pay their bills online at the click of a button. Electronic clearing system and electronic funds transfer facilitate faster funds movement and settlement for the customers of different banks and different centers. The electronic data interchange and cash management service facilities have enabled better Funds management for the customer. Very few banks offered customers the ability to access their accounts and perform at least simple money transactions using internet banking. Advancements in information technology had make possibility for the banks to use the internet as delivery channel for banking services. Technological

developments has introduced tremendous changes in the ability of financial and Non-Financial firms to efficiently collect, store, use and sell information about their customers.

Over a decade Indian banking system witnessed metamorphosis. The main driver of transformation has been the fast adoption of Information, Communication and Technology (ICT) based system in the banks. The huge red ledgers, row of racks of ledger holders, cash scrolls, registers, clearing cheque scrolls, totalling machines, long rolls of paper ribbons often gazing the floor formed part of hardware in the branches. It was also common to see staff hiding behind the tall branch counters, row of signature cabinets standing between the counters and supervisory staff, customers eyeing frantically on movement of ledgers and cheques until their transactions were done. But in the post bank reform era, more particularly after the ICT enablement there is semantic changes and innovation in the quality of customer services.

Moreover, the beeline of customers standing in queue in bank branches staring anxiously at the staff, their eagerness to catch up bank timings to log in transactions, searching for known employees to deposit/receive payments late at the counters, receiving wads of currency notes in retail payments at the counters, waiting for updating pass books, receiving drafts, grumbling over the bad handwriting of some of the employees were also the common features of 104 manual banking. They are now no more relevant. The banking work space has changed for good. Bank branches are now sporting a smart look with refurbished interior, radiating corporate colour, well dressed bank logos, wide glass doors, and plush interiors and well-developed customer lounges etc.

The well painted signage, clear guidance in the branch, customer information, display of product information, enquiry kiosk, smiling relationship assistants in

some banks adds to the modern branch set up. The low height counters handled by trained employees wearing inviting look, customers having one to one interface with departments, banking halls buzzing with clicks of mouse, laptops, computers, currency notes zipping through the counting machines form part of modernized attire of bank branches at least in metro cities. The eerie silence of customers and staff, an assured quick servicing system, provides an atmosphere for maintaining focused quality of service in the branches. The onsite ATMs, teller counters, swipe machines / kiosks have speed up standard transactions of every day need of consumers. With the onset of alternative delivery channels, even the branch timings are not very significant. Phone and mobile banking, smart cards, debit cards, rechargeable electronic purse are also some of the Modern day banking facilities that allow round the clock access. With the profile and aptitude of bank consumers fast changing toward the use of ICT facilities, the popularity of e-channels of banking are set to assume more significance. Banks are fast gearing up to introduce add-on services to attract young generation of customers. This connectivity has removed even the limitations in the use of debit/credit cards.

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The main channels for these were internet banking and mobile banking. After this, came the alliances for payment through various other gateways. The third important development happening now is the real-time gross settlement system of the RBI. Once this was in place, transactions between banks could be done through the settlement system, online, electronically thereby, ensuring faster collection. The process of computerization had started from Back Office Application, after that Total Branch Automation and nowadays it is the period of implementation of Core Banking Solutions (CBS). A key trend in the last couple of years has been the focus on core banking systems. With the implementation of core banking systems across the banks, the usage level of IT for customer management has increased. Core banking systems have enabled banks to launch new products and services targeting specific customer segments after understanding their banking and investment requirements. ATM, internet banking and mobile banking have improved customer convenience by providing anywhere any time banking services. The utility bill presenting and payment has helped customers to pay their bills

online at the click of a button. Electronic clearing system and electronic funds transfer have facilitated faster funds movement and settlement for the customers of different banks and different centres. The electronic data interchange and cash management service facilities have enabled better funds management for the customer.

Very few banks offered customers the ability to access their accounts and perform at least simple money transactions using internet banking. Advancements in information technology have made it possible for the banks to use the internet as a delivery channel for banking services. Technological developments have introduced tremendous changes in the ability of financial and non-financial firms to efficiently collect, store, use and sell information about their customers. Balasubramanya S.(2022) in his study analysed that the automation in the banking sector has come a long way starting with the Rangarajan Committee report on the banking sector reforms during the eighties, followed by reports of the Narasimhan Committee in the nineties. With over 65,000 branches of the banks (public, private and the cooperative sector) in the country, the author found that the percentage of branches covered by automation was very low. Though many banks had claimed that more than 70% business has been automated due to the enforcement of RBI guidelines, in reality it was much lower, as many functions in each branch were still done manually or with partial automation. Hence, there was a significant amount of automation work to be achieved in the banking sector.

Reserve Bank of India and Impact of Liberalization on Banking System

With liberalization in the telecom industry and its improved reliability at a reduced cost, many banks and financial sectors at that time were going forward with large-scale networking of their branches and implementing the centralized core

banking solutions. As a result, banks were able to provide their products and services to their customers anywhere, any time. With these developments, bank customers could avail these services across different locations with improved transaction realization and reduced cost. With increasing proliferation of ATMs, telebanking, and availability of internet banking facilities, the customer contact points had increased enormously, thereby resulting in increased services to customers. This has been possible solely due to the implementation of technology.

RBI has set up Department of Information Technology (DIT) which works for:

- Computerization in RBI (Regional Offices and Central Office Departments)
- Design and development of projects for use of banks and financial institutions and
- Monitoring progress of technology in banks

Current Focus of DIT:

i) Computerization in RBI DIT has been concentrating on computerization of all activities undertaken in the Banking Department (Deposit Accounts Department, Public Accounts Department, Public Debt Office, Establishment Section and Central Accounts Section) and the Issue Department (Currency Chest Management and Accounting) which impact on the balance sheet of the Reserve Bank. These departments also extend customer service. Computerization of these departments, therefore, aims at ensuring better housekeeping and efficient customer service.

ii) Design and Development of Projects for use of Banks and Financial Institutions
The projects developed so far and those listed for developments are as under:
Projects already developed:

- MICR cheque processing at four metros (Mumbai, New Delhi, Calcutta and Chennai) with image technology (July - October 1999)
- Electronic Clearing

Services (debit and credit) at 15 centers where RBI has its offices and 30 centers managed by SBI.

- Electronic Funds Transfer at four metros and its extension to Hyderabad, Ahmedabad and Bangalore
- Projects in the Process of Development:
- Indian Financial Network (INFINET)
- Securities Settlement System (SSS) and Negotiated Dealing System (NDS)
- Centralized Funds Management System (CFMS) • Structured Financial Messaging Solution (SFMS)
- Real Time Gross Settlement (RTGS)

(iii) Monitoring

- Progress in computerization and networking to achieve targets set by the Central Vigilance Commission of coverage of 70% of their business by computerization.
- Setting up MICR Cheque Processing centers at non-metros • Adoption of standardization in the area of hardware, operating system and communication platforms • Development of generic architecture e (tree or star topology for domestic and cross border connectivity)

IT Framework for Indian Banking sector

IT planning is an ongoing effort intended to match the bank's technology capabilities with its changing strategic objectives. It is necessary for a bank to identify technology gaps and develop a plan that supports the bank's long/medium term-strategic goals in order to bridge the gaps. It is imperative for banks to have a clearly defined technology planning process that is based on a well founded technology action plan for the following reasons:

- Increasing competition, new products and changing distribution channels.
- Banks currently spend a huge amount of their budget annually on

technology. Such investments will only continue to escalate.

- Effective technology management requires an underlying technology plan. Without it, scarce resources are likely to be wasted and opportunities missed.

Gulati et al. (2023) suggested IT policy framework for Indian banks as follows. IT strategies need to be formulated by banks taking into consideration the critical aspects of long/short-term planning to align technology systems with business objectives. Conscious efforts must be made to place the entire organization's proper perspective and to have a holistic approach to planning. The following strategic evaluation needs to be made:

- Current state (Where are we?): There should be a self-assessment process which analyses the present/current technology in use. It also involves evaluation of staffing, training, organizational processes and controls, communication and management reporting. To successfully integrate new technologies, banks must objectively confront internal operating issues and be willing to make changes wherever necessary. Business process re-engineering should be accorded top priority to successfully absorb new technology.
- Desired state (Where do we want to go?): Identification and prioritization of the reasons behind technology adoption is vital. Technology goals should always be firmly grounded in an understanding of the marketplace. Sizing up the competition and measuring up to its pace, based on a SWOT analysis, must be the foundation of the decision on where to go.
- Destination (How do we get there?): This phase of the technology planning process, involves making decisions about, how to implement the technology action plan and the technology initiatives required to be pursued in the short/mid/long-term.

As part of the planning of technology initiatives, a list of projects to be undertaken needs to be made. For this, the element of time span should be considered relative to the bank's position and future needs (what initiatives are planned in short/mid/long term). A technology plan is a document that lays down the steps necessary for each action item. It serves as a road map for investment.

Role of ICT in Banking

Technology is no longer being used simply as a means for automating processes. Instead it is being used as a revolutionary means of delivering services to customers. The adoption of technology has led to the following benefits: greater productivity, profitability, and efficiency; faster service and customer satisfaction; convenience and flexibility; 24x7 operations; and space and cost savings (Sivakumaran, 20021). Harrison Jr., chairman and chief executive officer of Chase Manhattan, which pioneered many innovative applications of ICT in banking industry, observed that the Internet caused a technology revolution and it could have greater impact on change than the industrial revolution (Engler&Essinger, 2020). Technology has been used to offer banking services in the following ways (Sivakumaran, 2023):

ATM:



- ATMs are the cash dispensing machines that can be seen at banks and other locations where crowd proximity is more. ATMs started as a substitute to a bank to allow its customers to withdraw cash at any time and to provide services where it would not be viable to open another physical branch. The ATM is the most visited delivery channel in retail banking, with more than 40 billion transactions annually worldwide. In fact, the delivery channel revolution is said to have begun with the ATM. It was indeed a pleasant change for customers to be in charge of their transaction, as no longer would they need to depend on an indifferent bank employee. ATMs have made banks realize that they could divert the huge branch traffic to the ATM. The benefits hence were mutual. Once banks realized the convenience of ATMs, new services started to be added.
- The phenomenal success of ATMs had made the banking sector develop more innovative delivery channels to build on cost and service efficiencies. As a consequence, banks have introduced telebanking, call centers, Internet banking, and mobile banking. Telebanking is a good medium for customers

to make routine queries and also an efficient tool for banks to cut down on their manpower resources. The is call center another channel that captured the imagination of banks as well as customers. At these centers, enormous amount of information is at the fingertips of trained customer service representatives. A call center meets a bank's infrastructural, as well as customer service requirements. Not only does a call center cut down on costs, it also results in customer satisfaction. Moreover, it facilitates 24x7 working and offers the "human touch" that customers seek. The call center has large potential dividends by way of improved customer relationship management (CRM) and return on investment (ROI).

- With the Internet boom, banks realized that Internet banking would be a good way to reach out to customers. Currently, some banks are attempting to harness the benefits of Internet banking, while others have already made Internet banking an important and popular payment system. Internet banking is on the rise, as is evident from the statistics. Predictions of Internet banking To go the ATM way have not materialized as much as anticipated; many reasons can be cited for this. During 2003, the usage of the Internet as a banking channel accounted for 8.5%. But this was due to the false, unrealistic expectations tied to it. Some of the factors that were detrimental in bringing down, or rather, not being supportive, are low Internet penetration, high telecom tariffs, slow Internet speed and inadequate bandwidth availability, lack of extended applications, and lack of a trusted environment.
- Before an ATM is placed in a public place, it typically has undergone extensive testing with both test money and the backend computer systems that allow it to perform transactions. Banking customers also have come to expect high reliability in their ATMs, which provides incentives to ATM providers to minimize machine and network failures. Financial consequences of incorrect machine operation also provide high degrees of incentive to minimize malfunctions

- ATMs and the supporting electronic financial networks are generally very reliable, with industry benchmarks typically producing 98.25% customer availability for ATM sand up to 99.999% availability for host systems that manage the networks of ATMs. If ATM networks do go out of service, customers could be left without the ability to make transactions until the beginning of their bank's next time of opening hours. To aid in reliability, some ATMs print each transaction to a roll-paper journal that is stored inside the ATM, which allows its users and the related financial institutions to settle things based on the records in the journal in case there is a dispute.
- A payment terminal, also known as a Point of Sale (POS) terminal, credit card terminal, EFTPOS terminal (or by the older term as PDQ terminal which stands for "Process Data Quickly "or in common jargon as "Pretty Damn Quick" is a device which interfaces with payment cards to make electronic fund transfer, The terminal typically consists of a secure keypad (called a PIN pad) for entering PIN, a screen, a means of capturing information from payments cards and a network connection to access the payment network for authorization.
- A payment terminal allows a merchant to capture required credit and debit card information and to transmit this data to the merchant services provider or bank for authorization and finally, to transfer funds to the merchant. The terminal allows the merchant or their client to swipe, insert or hold a card near the device to capture the information. Terminal is often connected to point of sale systems so that payment amounts and confirmation of payment can be transferred automatically to the merchant's retail management system. Terminals can also be used in standalone mode, where the merchant keys the amount into the terminal before the customer present their card and personal identification number (PIN)

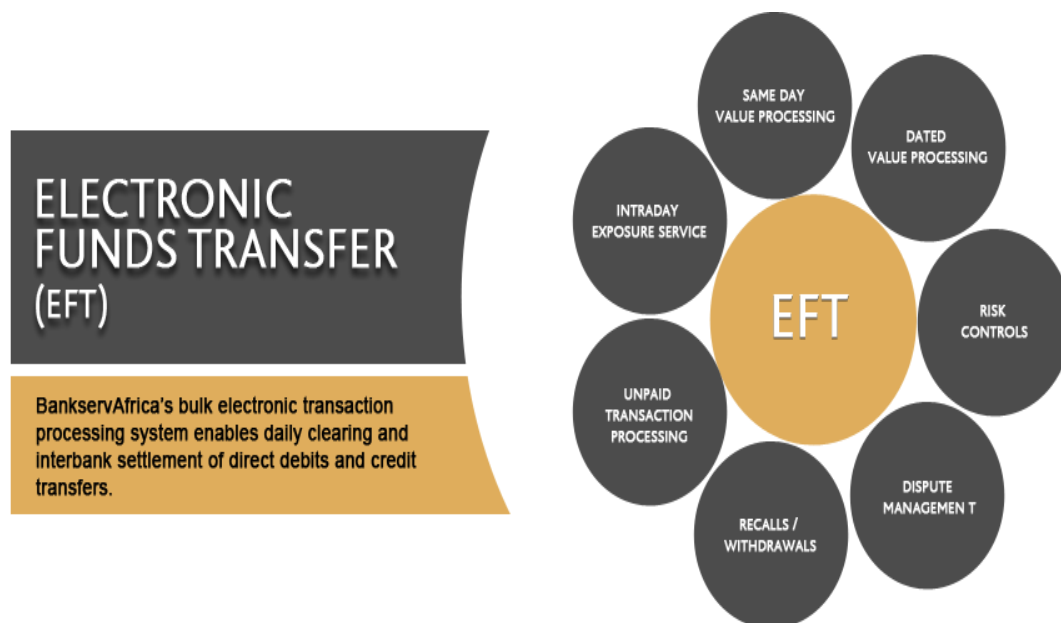
- Payment cards are part of payment system issued financial institutional, such as a bank, to a customer that enables its owner (the cardholder) to access the funds in the customer's designated bank accounts, or through a credit accounts and make payments by electronic fund transfer and access automated teller machine (ATMs). Such cards are known by a variety of names including bank cards, ATM cards, MAC (money access cards), client cards, key cards or cash cards.



Electronic fund transfer

.Electronic funds transfer (EFT) are electronic transfer of money from one bank account to another, either within a single financial institution or across multiple institutions, via computer-based systems, without the direct intervention of bank staff. Electronic Funds Transfer (EFT) is a system of transferring money from one bank account directly to another without any paper money changing hands. One of the most widely-used EFT programs is Direct Deposit, in which payroll is deposited straight into an employee's bank account, although EFT refers to any transfer of funds initiated through an electronic terminal, including credit card, ATM, Fed wire and point-of-sale (POS) transactions. It is used for both credit transfers, such as payroll payments, and for debit transfers, such as mortgage payments. Transactions are processed by the bank through the Automated Clearing House (ACH) network, the secure transfer system that connects all U.S. financial institutions. For payments, funds are transferred electronically from one bank

account to the billing company's bank, usually less than a day after the scheduled payment date



The Electronic Fund Transfer Act (EFTA) (15 USC 1693 et seq.) of 1978 is intended to protect individual consumers engaging in electronic fund transfers (EFTs). EFT services include transfers through automated teller machines, point-of-sale terminals, automated clearinghouse systems, telephone bill-payment plans in which periodic or recurring transfers are contemplated, and remote banking programs. The Federal Reserve Board (Board) implements EFTA through Regulation E, which includes an official staff commentary. The Electronic Signatures in Global and National Commerce Act (the E-Sign Act), 15 USC 7001 et seq., became effective October 1, 2000, and allows electronic documents and signatures to have the same validity as paper documents and handwritten signatures. Disclosures in consumer transactions provided in electronic form would satisfy Regulation E's written disclosure requirement only if the financial institution received proper consent under the E-Sign Act. If a financial institution provides disclosures in both paper and electronic form, the paper form can be used to meet the disclosure requirements, and E-Sign consent is not required. The Board issued

final rules for the electronic delivery of disclosures required under Regulation E on December 10, 2007 (72 Fed. Reg. 63,452 (Nov. 9, 2007)).

National Electronic Funds Transfer (NEFT)

National Electronic Funds Transfer (NEFT) is an electronic funds transfer system maintained by the Reserve Bank of India (RBI). Started in November 2005, the setup was established and maintained by Institute for Development and Research in Banking Technology (IDRBT). NEFT is a facility enabling bank customers in India to transfer funds between any two NEFT-enabled bank accounts on a one-to-one basis. It is done via electronic messages. Unlike Real-time gross settlement (RTGS), fund transfers through the NEFT system do not occur in real-time basis. NEFT settles fund transfers in half-hourly batches with 23 settlements occurring between 8:00 AM and 7:00 PM on week days and the 1st, 3rd and 5th Saturday of the calendar month. Transfers initiated outside this time period are settled at the next available window. No settlements are made on the second and fourth Saturday of the month, or on Sundays, or on public holidays.

NEFT facilities are available at 74,680 branches offices of 101 banks across the country (out of around 82,400 bank branches) as of January 2011, and well as online through the website of NEFT-enabled banks and work on a batch mode. NEFT has gained popularity due to its saving on time and the ease with which the transactions can be concluded, This reflects from the fact that 42% of all electronic transactions in the 2008 financial year were NEFT transactions.

Detailed process NEFT is as follows:

Indian Bank customer fills an application form providing details of the beneficiary (like name, bank, branch name, IFSC, account type and account number) and the amount to be remitted. The remitter authorizes his/her bank branch to debit his account and remit the specified amount to the beneficiary. This facility

is also available through online banking and some banks offer the NEFT facility even through the ATMs.

1. The originating bank branch prepares a message and sends the message to its pooling center (also called the NEFT Service Centre).
2. The pooling center forwards the message to the NEFT Clearing Centre (operated by National Clearing Cell, Reserve Bank of India, Mumbai) to be included for the next available batch.
3. The Clearing Centre sorts the funds transfer transactions destination bank-wise and prepares accounting entries to receive funds from the originating banks (debit) and give the funds to the destination banks(credit). Thereafter, bank-wise remittance messages are forwarded to the destination banks through their pooling centre (NEFT Service Centre).
4. The destination banks receive the inward remittance messages from the Clearing Centre and pass on the credit to the beneficiary customers' accounts.

Real-time gross settlement (RTGS)

Real-time gross settlement (RTGS) systems are specialist funds transfer systems where the transfer of money or securities takes place from one bank to any other bank on a "real time" and on a "gross" basis. Settlement in "real time" means a payment transaction is not subjected to any waiting period, with transactions being settled as soon as they are processed. "Gross settlement" means the transaction is settled on one-to-one basis without bundling or netting with any other transaction. "Settlement" means that once processed, payments are final and irrevocable.

RTGS systems are typically used for high-value transactions that require and receive immediate clearing. In some countries the RTGS systems may be the only way to get same day cleared funds and so may be used when payments need to be settled urgently. However, most regular payments would not use a RTGS system,

but instead would use a national payment system or automated clearing house that allows participants to batch and net payments. RTGS payments typically incur higher transaction costs and usually operated by a country's central bank.

RTGS systems are usually operated by a country's central bank as it is seen as a critical infrastructure for a country's economy. Economists believe that an efficient national payment system reduces the cost of exchanging goods and services, and is indispensable to the functioning of the interbank, money, and capital markets. A weak payment system may severely drag on the stability and developmental capacity of a national economy; its failures can result in inefficient use of financial resources, inequitable risk-sharing among agents, actual losses for participants, and loss of confidence in the financial system and in the very use of money.

RTGS system does not require any physical exchange of money; the central bank makes adjustments in the electronic accounts of Bank A and Bank B, reducing the balance in Bank A's account by the amount in question and increasing the balance of Bank B's account by the same amount. The RTGS system is suited for low-volume, high-value transactions. It lowers settlement risk, besides giving an accurate picture of an institution's account at any point of time. The objective of RTGS systems by central banks throughout the world is to minimize risk in high-value electronic payment settlement systems. In an RTGS system, transactions are settled across accounts held at a central bank on a continuous gross basis. Settlement is immediate, final and irrevocable. Credit risks due to settlement lags are eliminated. The best RTGS national payment system cover up to 95% of high-value transactions within the national monetary market.

RTGS vs NEFT

Real Time Gross Settlement	National Electronics Funds Transfer
• Real-time settlement	• Deferred net settlement basis
• Funds transfers at the time they are received	• Transfer after a cut-off time
• Settlement of funds transfer instructions occurs individually	• Operates in hourly batches
• Processed continuously throughout the business hours.	• Eleven settlements from 9 am to 7 pm on weekdays
• Meant for Rs. 2 lakhs and above	• Five settlements from 9 am to 1 pm on Saturdays
• 9.00 to 16.30 on weekdays. 9.00 to 13.30 hours on Saturdays.	• After this you will have to wait till the next designated time

Online banking

Online banking, also known as internet banking, is an electronic payment system that enables customers of a bank or other financial institution to conduct a range of financial transactions through the financial institution's website. The online banking system will typically connect to or be part of the core banking system operated by a bank and is in contrast to branch banking which was the traditional way customers accessed banking services.

Some banks operate as a "direct bank" (or “virtual bank”), where they rely completely on internet banking.

Internet banking software provides personal and corporate banking services offering features such as viewing account balances, obtaining statements, checking recent transaction and making payments. Access is usually through a secure web site using a username and password, but security is a key consideration in internet banking and many banks also offer two factor authentications using a (security token).

Operations; To access a financial institution's online banking facility, a customer with internet access will need to register with the institution for the service, and set up a password and other credentials for customer verification. The credentials for

online banking is normally not the same as for telephone or mobile banking. Financial institutions now routinely allocate customers numbers, whether or not customers have indicated an intention to access their online banking facility. Customer numbers are normally not the same as account numbers, because a number of customer accounts can be linked to the one customer number. Technically, the customer number can be linked to any account with the financial institution that the customer controls, though the financial institution may limit the range of accounts that may be accessed to, say, cheque, savings, loan, credit card and similar accounts.

The customer visits the financial institution's secure website, and enters the online banking facility using the customer number and credentials previously set up. Each financial institution can determine the types of financial transactions which a customer may transact through online banking, but usually includes obtaining account balances, a list of recent transactions, electronic bill payments, financing loans and funds transfers between a customer's or another's accounts. Most banks set limits on the amounts that may be transacted, and other restrictions. Most banks also enable customers to download copies of bank statements, which can be printed at the customer's premises (some banks charge a fee for mailing hard copies of bank statements). Some banks also enable customers to download transactions directly into the customer's accounting software. The facility may also enable the customer to order a cheque book, statements, report loss of credit cards, stop payment on a cheque, advice change of address and other routine actions.

- **Features**

Online banking facilities typically have many features and capabilities in common, but also have some that are application specific. The common features fall broadly into several categories:

1. A bank customer can perform non-transactional tasks through online banking, including:
 - Viewing account balances
 - Viewing recent transactions
 - Downloading bank statements, for example in PDF format
 - Viewing images of paid cheques
 - Ordering cheque books
 - Download periodic account statements
 - Downloading applications for M-banking, E-banking etc.
2. Bank customers can transact banking tasks through online banking, including:
 - Funds transfers between the customer's linked accounts
 - Paying third parties, including bill payments (see, e.g., BPAY) and third party fund transfers (see, e.g., FAST)
 - Investment purchase or sale
 - Loan applications and transactions, such as repayments of enrollments
 - Credit card applications
 - Register utility billers and make bill payments
3. Financial institution administration
4. Management of multiple users having varying levels of authority
5. Transaction approval process



- **Security**

Security of a customer's financial information is very important, without which online banking could not operate. Similarly, the reputational risks to banks themselves are important. Financial institutions have set up various security processes to reduce the risk of unauthorized online access to a customer's records, but there is no consistency to the various approaches adopted.

The use of a secure website has been almost universally embraced.

Though single password authentication is still in use, it by itself is not considered secure enough for online banking in some countries. Basically, there are two different security methods in use for online banking:

- The PIN/TAN system where the PIN represents a password, used for the login and TANs representing one-time passwords to authenticate transactions. TANs can be distributed in different ways; the most popular one is to send a list of TANs to the online banking user by postal letter. Another way of using TANs is to generate them by need using a security token. These token generated TANs depend on the time and a unique secret, stored in the security token (two-factor authentication or 2FA).
- More advanced TAN generators (chip TAN) also include the transaction data into the TAN generation process after displaying it on their own screen to allow

the user to discover man-in-the-middle attacks carried out by Trojans trying to secretly manipulate the transaction data in the background of the PC.

- Another way to provide TANs to an online banking user is to send the TAN of the current bank transaction to the user's (GSM) mobile phone via SMS. The SMS text usually quotes the transaction amount and details; the TAN is only valid for a short period of time. Especially in Germany, Austria and the Netherlands many banks have adopted this "SMS TAN" service.
- Usually online banking with PIN/TAN is done via a web browser using SSL secured connections, so that there is no additional encryption needed.
- Signature based online banking where all transactions are signed and encrypted digitally. The Keys for the signature generation and encryption can be stored on smartcards or any memory medium.^l



- **Attacks;** Attacks on online banking used today are based on deceiving the user to steal login data and valid TANs. Two well-known examples for those attacks are phishing and pharming. Cross-site scripting and key logger/Trojan horses can also be used to steal login information.

A method to attack signature based online banking methods is to manipulate the used software in a way, that correct transactions are shown on the screen and faked transactions are signed in the background.

A 2008 U.S. Federal Deposit Insurance Corporation Technology Incident Report, compiled from suspicious activity reports banks file quarterly, lists 536 cases of computer

intrusion, with an average loss per incident of \$30,000. That adds up to a nearly \$16-million loss in the second quarter of 2007. Computer intrusions increased by 150 percent between the first quarter of 2007 and the second. In 80 percent of the cases, the source of the intrusion is unknown but it occurred during online banking, the report states.

Another kind of attack is the so-called man-in-the-browser attack, a variation of the man-in-the-middle attack where a Trojan horse permits a remote attacker to secretly modify the destination account number and also the amount in the web browser. As a reaction to advanced security processes allowing the user to cross-check the transaction data on a secure device there are also combined attacks using malware and social engineering to persuade the user himself to transfer money to the fraudsters on the ground of false claims (like the claim the bank would require a "test transfer" or the claim a company had falsely transferred money to the user's account and he should "send it back"). Users should therefore never perform bank transfers they have not initiated themselves.

- **Countermeasure**

There exist several countermeasures which try to avoid attacks. Digital certificates are used against phishing and pharming, in signature based online banking variants (HBCI/FinTS) the use of "Decoder" card readers is a measurement to uncover software side manipulations of the transaction data.

In 2001, the U.S. Federal Financial Institutions Examination Council issued guidance for multifactor authentication (MFA) and then required to be in place by the end of 2006.

In 2012, the European Union Agency for Network and Information Security advised all banks to consider the PC systems of their users being infected by malware by default and therefore use security processes where the user can cross-check the transaction data against manipulations like for example (provided the security of the mobile phone holds up) SMS TAN where the transaction data is sent along with the TAN number or standalone smartcard readers with an own screen including the

transaction data into the TAN generation process while displaying it beforehand to the user (see chipTAN) to counter man-in-the-middle attacks.

- **Mobile Banking**

Mobile banking is a service provided by a bank or other financial institution that allows its customers to conduct financial transactions remotely using a mobile device such as a smartphone or tablet. Unlike the related internet banking it uses software, usually called an app, provided by the financial institution for the purpose. Mobile banking is usually available on a 24-hour basis. Some financial institutions have restrictions on which accounts may be accessed through mobile banking, as well as a limit on the amount that can be transacted. Mobile banking is dependent on the availability of an internet or data connection to the mobile device.

Transactions through mobile banking depend on the features of the mobile banking app provided and typically includes obtaining account balances and lists of latest transactions, electronic bill payments, remote check deposits, P2P payments, and funds transfers between a customer's or another's accounts. Some apps also enable copies of statements to be downloaded and sometimes printed at the customer's premises.

From the bank's point of view, mobile banking reduces the cost of handling transactions by reducing the need for customers to visit a bank branch for non-cash withdrawal and deposit transactions. Mobile banking does not handle transactions involving cash, and a customer needs to visit an ATM or bank branch for cash withdrawals or deposits. Many apps now have a remote deposit option; using the device's camera to digitally transmit cheques to their financial institution. Mobile banking differs from mobile payments, which involves the use of a mobile device to pay for goods or services at the point of sale or remotely, analogously to the use of a debit or credit card to affect an EFTPOS payment. Mobile Banking refers to provision and availability of banking- and financial services with the help of mobile

telecommunication devices. The scope of offered services may include facilities to conduct bank and stock market transactions, to administer accounts and to access customized information."



According to this model mobile banking can be said to consist of three inter-related concepts:

- Mobile accounting
- Mobile brokerage
- Mobile financial information services

Most services in the categories designated accounting and brokerage are transaction-based. The non-transaction-based services of an informational nature are however essential for conducting transactions - for instance, balance inquiries might be needed before committing a money remittance. The accounting and brokerage services are therefore offered invariably in combination with information services. Information services, on the other hand, may be offered as an independent module.

Mobile banking may also be used to help in business situations as well as financial

- **Account Information**

1. Mini-statements and checking of account history
2. Alerts on account activity or passing of set thresholds
3. Monitoring of term deposits
4. Access to loan statements
5. Access to card statements
6. Mutual funds / equity statements
7. Insurance policy management

- **Transaction**

1. Funds transfers between the customer's linked accounts
2. Paying third parties, including bill payments and third party fund transfers
3. Check Remote Deposit

- **Investment**

1. Portfolio management services
2. Real-time stock

- **Support**

1. Status of requests for credit, including mortgage approval, and insurance coverage
2. Check (cheque) book and card requests
3. Exchange of data messages and email, including complaint submission and tracking
4. ATM Location

- **Content Services**

1. General information such as weather updates, news
2. Loyalty-related offers
3. Location-based services

A report by the US Federal Reserve (March 2012) found that 21 percent of mobile phone owners had used mobile banking in the past 12 months. Based on a survey conducted by Forrester, mobile banking will be attractive mainly to the younger, more "tech-savvy" customer segment. A third of mobile phone users say that they may consider performing some kind of financial transaction through their mobile phone. But most of the users are interested in performing basic transactions such as querying for account balance and making bill.

- **Challenges for a Mobile Banking Solution**

There are a large number of different mobile phone devices and it is a big challenge for banks to offer a mobile banking solution on any type of device. Some of these devices support Java ME and others support SIM Application Toolkit, a WAP browser, or only SMS.

Initial interoperability issues however have been localized, with countries like India using portals like "R-World" to enable the limitations of low-end java-based phones, while focus on areas such as South Africa have defaulted to the USSD as a basis of communication achievable with any phone.

The desire for interoperability is largely dependent on the banks themselves, where installed applications (Java based or native) provide better security, are easier to use and allow development of more complex capabilities similar to those of internet banking while SMS can provide the basics but becomes difficult to operate with more complex transactions.

There is a myth that there is a challenge of interoperability between mobile banking applications due to perceived lack of common technology standards for mobile banking. In practice it is too early in the service lifecycle for interoperability to be addressed within an individual country, as very few countries have more than one mobile banking service provider. In practice, banking interfaces are well defined and money movements between banks follow the ISO-8583 standard. As mobile banking matures, money movements between service providers will naturally adopt the same standards as in the banking world.

In January 2009, Mobile Marketing Association (MMA) Banking Sub-Committee, chaired by Cell Trust and VeriSign Inc., published the Mobile Banking Overview for financial institutions in which it discussed the advantages and disadvantages of Mobile Channel Platforms such as Short Message Services (SMS), Mobile Web, Mobile Client Applications, SMS with Mobile Web and Secure SMS

- **CYBER SECURITY IN BANKING INDUSTRY**

There is a noticeable shift in the banking industry in the way customers deal with their transactions. There is a rapid increase in the usage of digital channels such as internet banking, digital wallets, mobile banking, ATM. This leads to the increase in exposure and thereby cyber attacks which further may lead to financial and reputational losses. Banks may lose the customer confidence which can further increase the impact. The key influencers who make it imperative for the banks to invest in security are: Increase in financial data losses including card data, personal identifiable information etc. Unauthorized access to bank's network and systems.

Key drivers for investment in Cyber Security:



Secure sensitive customer information



Fortify IT Processes and Systems



Customer demands for convenience and payment security



Adhere to regulatory requirements

With increasing risks of cyber threats, banks are facing an unprecedented challenge of data breaches and are therefore strengthening their cyber security postures. The following are the noticeable trends in banking industry from cyber security point of view: Financial sector faced almost three times the cyber attacks as compared to that of the other industries. Data breaches (both internal through fraud and external through cyber criminals) leads to the exponential rise in costs. It has been estimated that cost of implementing and managing the cyber security infrastructure will increase over 40% by 2025. There is an increase in biometrics and tokenization as banks have begun to recognize that in addition to being a solution for payments these controls are also useful in security the sensitive data. Customers are using biometrics for banking, activities such as authentication for mobile banking, transaction at ATMs and payments with digital channels becoming the preference. Choice of customers for banking services, banks will also need to leverage advanced authentication and access control processes, without any compromise to customer experience.



With the increase in the development of technologies the banking industry is evolving at an extraordinary rate. Unmanned aerial systems, the Internet of Things, Near Field of Communication (NFCs), and nearable devices are some of the technological advancements that banks will need to consider in the near future. Few of the top upcoming priorities for banks could be cloud based platforms, robotic process automation and cognitive technologies. Automation will drive new efficiencies across the security lifecycle, but require the creation of control mechanisms and strong governance. The above trends however pose their own set of challenges which are discussed in the next section.

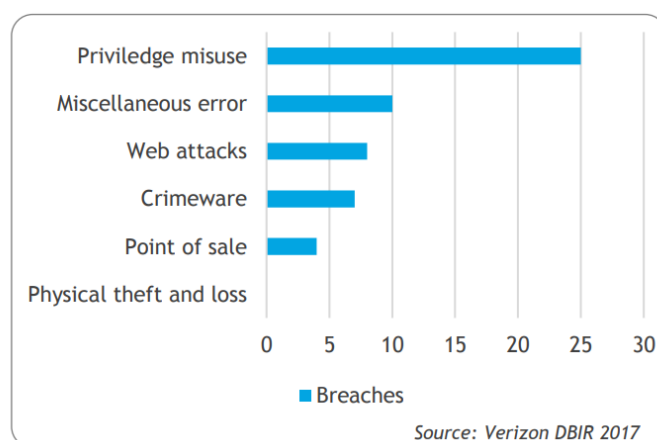
CHALLENGES

The exponential growth of digital payments platform in India and the push towards a cashless economy has renewed focus on the need to strengthen cybersecurity posture. Few of the major challenges faced by banks include:

- Strict compliance regulations: Managing regulatory compliances has become enormously challenging for the banks. Over the past few years the

volume of regulations has increased dramatically. Along with the larger banks, smaller ones too are required to fulfill the regulatory obligations. The struggle to secure.

- The struggle to secure customer data: There are number of ways in which violation of privacy can take place in banking sector like stolen or loss card data, unauthorized sharing of data with third parties and loss of client's personal data due to improper security measures
- Third party risk: Banks need to conduct due diligence on third parties they are associated with. As per Payments card industry data security standard, third parties need to report any critical issues associated the card data environment to the bank .
- Evolving cyber threat landscape: The development in technologies is leading to the latest cyber threats like next generation ransomwares, web attacks etc
- Transaction frauds: Fraud detection technologies should be in place with proper consideration of risks based on the business factors.
- Secure SDLC: Banks need to incorporate SDLC security for banking products and applications.



REGULATORY PERSPECTIVE:

To ensure security in banking industries, the Reserve Bank of India removed a Circular DBS.CO.ITC.BC.No.6/31.02.008/ 2010-11 dated April 29 2011, where all banking institutions have to comply for. Some of the key features of the regulations are

- Cyber Security Policy to be distinct from the broader IT policy / IS Security Policy of a bank
- Arrangement for continuous surveillance
- Comprehensive network and database security
- Protection of customer information
- Cyber security preparedness indicators
- Cyber Crisis Management Plan
- IT architecture should be conducive to security
- An immediate assessment of gaps in preparedness to be reported to RBI
- Cyber security awareness among stakeholders/ Top Management

SECURITY CONSIDERATIONS:

While each bank thinks distinctively on adopting various considerations it is imperative to assume that the theme remains the same for various banking channels:



Internet Banking: Security controls like multi factor authentication, creation of strong passwords, adaptive authentication, image authentication, etc. can be considered.



Mobile Banking: It should be ensured that mobile applications are up to date and should be tested. Latest hardening standards could be implemented.



Wallet Transactions: Awareness material on Phishing, Malware attacks, vishing and social engineering, Password security etc. should be incorporated.



ATM Security: Biometrics like eye-retina, voice scan or fingerprint scan should be introduced by Banks.



UPI (Unified Payment Interface) : Banks and PSPs need to think through their security strategies, governance models and predictive controls to build a secure UPI environment that ensures a seamless user experience and at the same time balances security risks.



Banks must conduct regular drills, awareness programs and simulation exercises to keep their infrastructure secured.

APPROACH TO SECURITY:

At BDO India, we Endeavour to provide expertise driven solutions to help and assist our clients business needs are met, through a well defined risk based approach Approach to adoption can have the following phases:

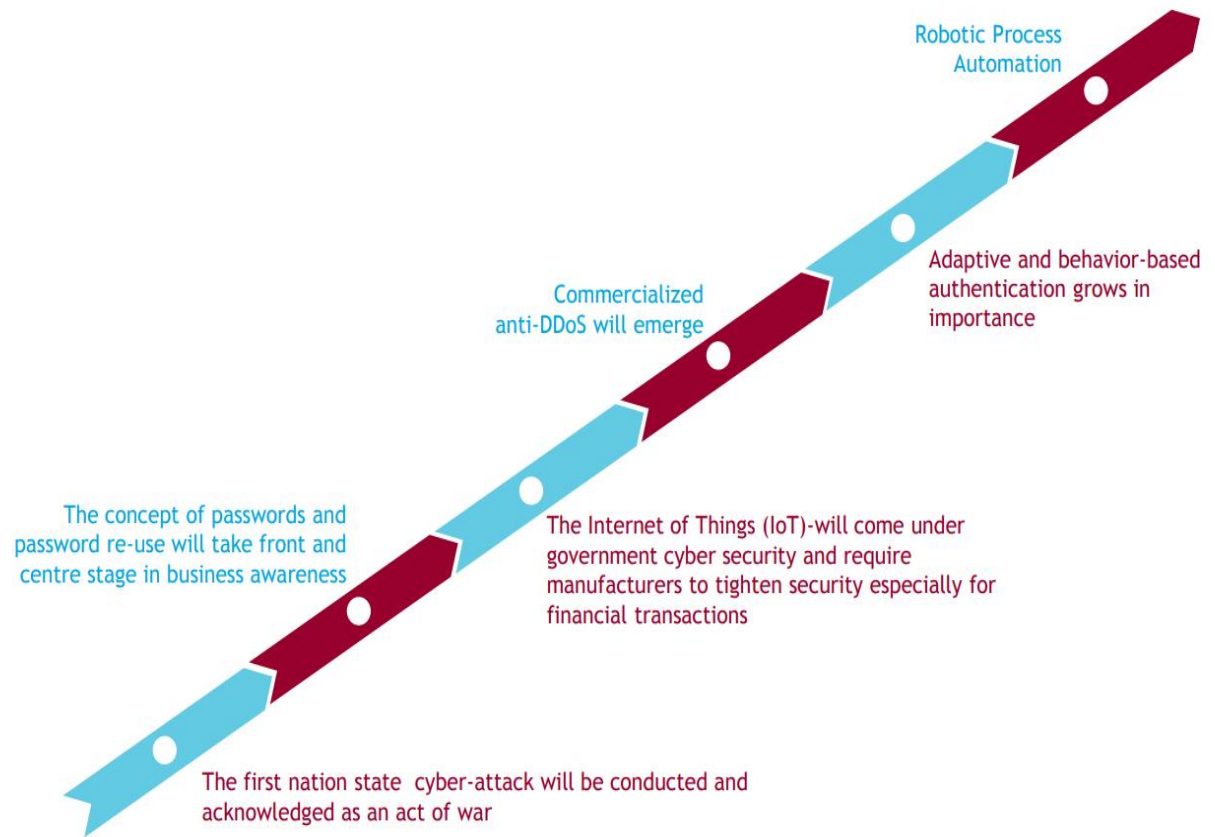
- **Plan:** Discussing the scope of work, making a roadmap for the approach, formalizing leadership & project SPOC and to understand the policy and procedures all can be a part of the planning phase.
- **Build & Design:** The build phase consists of requirements as a part of a systems engineering process. The main milestone of design phase would be matching the system specifications and the disposition of risk from the organization as shown in the framework.
- **Implementation:** Gaps identified during the plan phase are implemented. Integration elements should be carefully planned.

- Transition: A seamless transition and handover to the operations team should be taken into consideration.
- Manage: This phase includes management, monitoring, and periodic reviews against security threats and frauds.



CYBER SECURITY TRENDS:

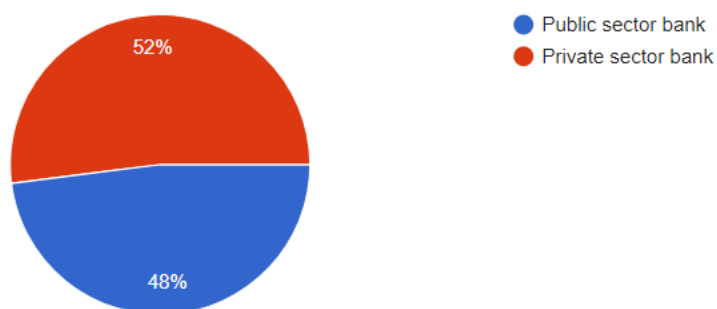
- Blockchain is a technology that was initially developed for Bitcoin, the cryptocurrency. Blockchain could reduce banks infrastructure costs by US\$ 15-20 billion per annum by 2022. Blockchain have the potential to transform how the business and the government work in vast variety of contexts.
- Banks will continue to leverage digital technologies to enhance customer experience.
- Ongoing threats related to IoT devices will force banks to tighten security layers, including patchable firmware/software, secured authentication, and controlled privilege access. Today, most IoT devices are considered throw away devices and security patches are not issued. But, new regulations will be driven by large scale attacks using IoT to amplify the attack.



CHAPTER 5

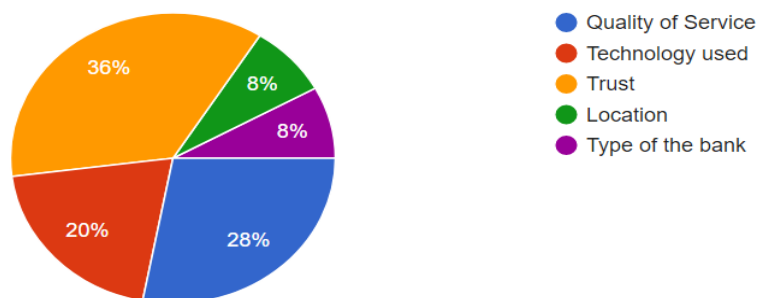
DATA ANALYSIS AND INTERPRETATION

1. Which category of the banks do you consider as most technologically advanced?



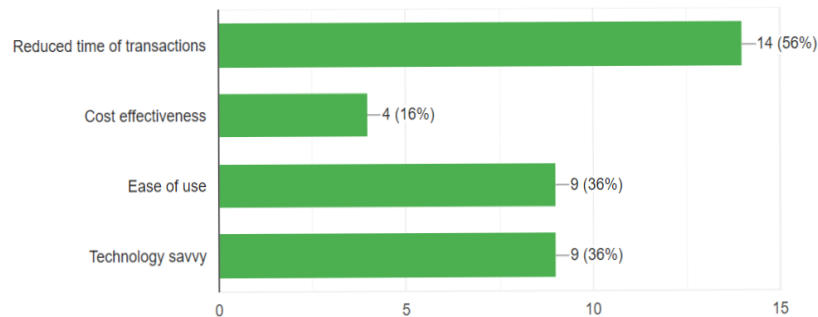
Ans. There are at most 52% in private sector banks, whereas, there are 48% in Private sector banks

2. Which attribute of the bank do you value the most?



Ans. People always believe in trustworthy banks ,who always stay loyal with their customers, whereas some peoples also go for quality of service given by banking sector.

3. Which factor promotes you to use the new techniques in banking?

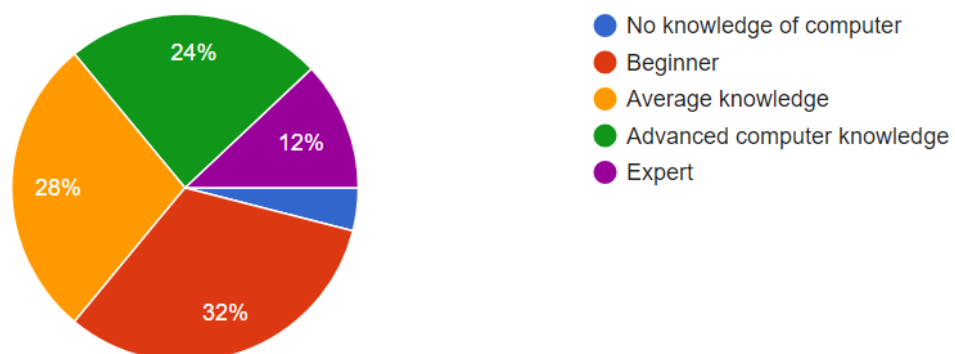


Ans.

People always used to go for easier way of transaction in banking sector, if there is a technological development in market then it will promotes new techniques in banking sector.

4. How familiar are you with computer usage level of your bank?

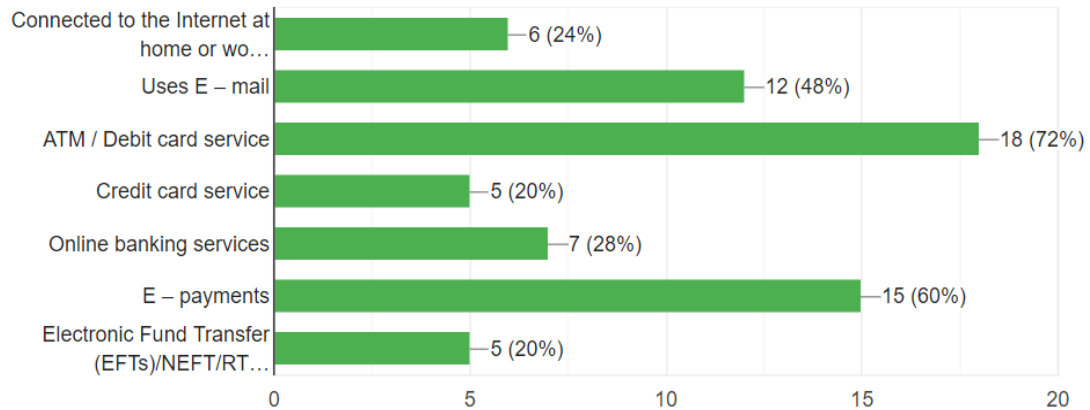
Ans.



There should be more awareness of using computer techniques in banking sector, thus this helps to grow technological development in banks

5. Customer level of usage of technology.

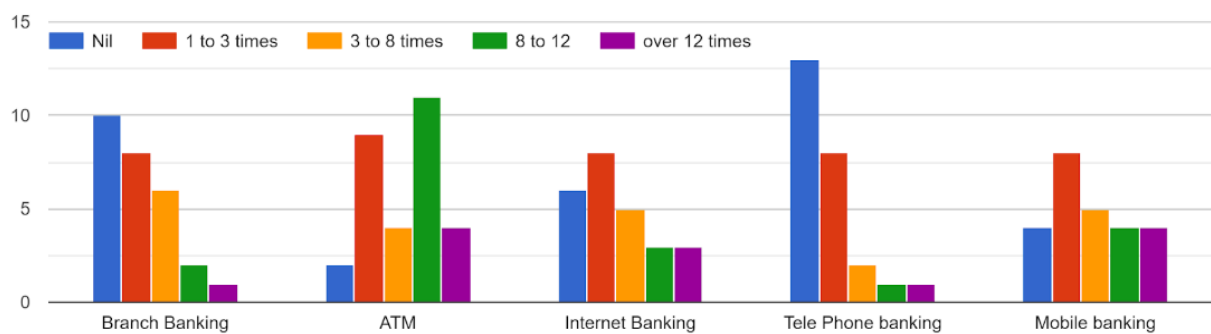
Ans



The most used technology in Banking sector are ATM services, whereas there is also an huge demand in E-payment where they can use easily withdraw cash.

6. How frequently do you use the following Indian banking services per month?

Ans

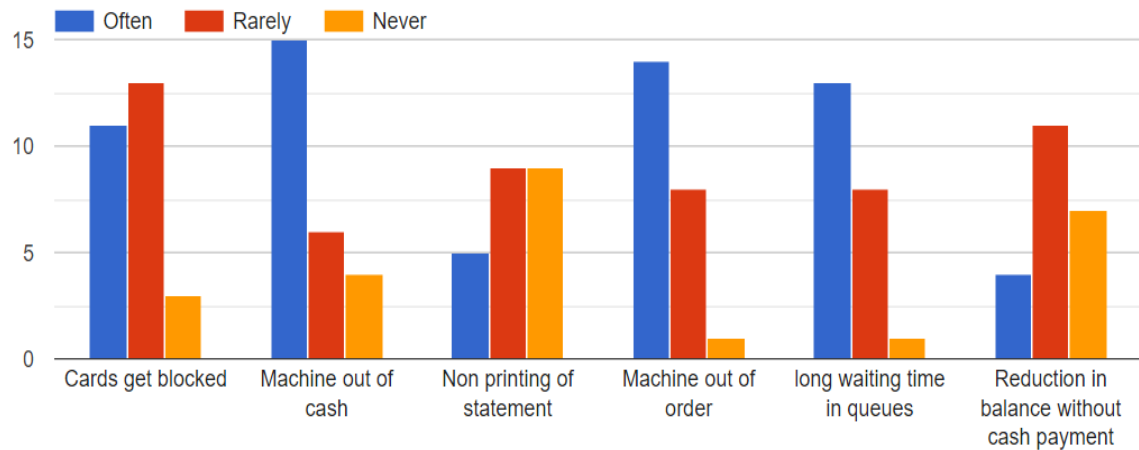


As the technological development is increasing day by day there is a lesser demand for Tele-phone banking, whereas ATM services are most used by customers in Banking sectors

Problems of Technology Usage

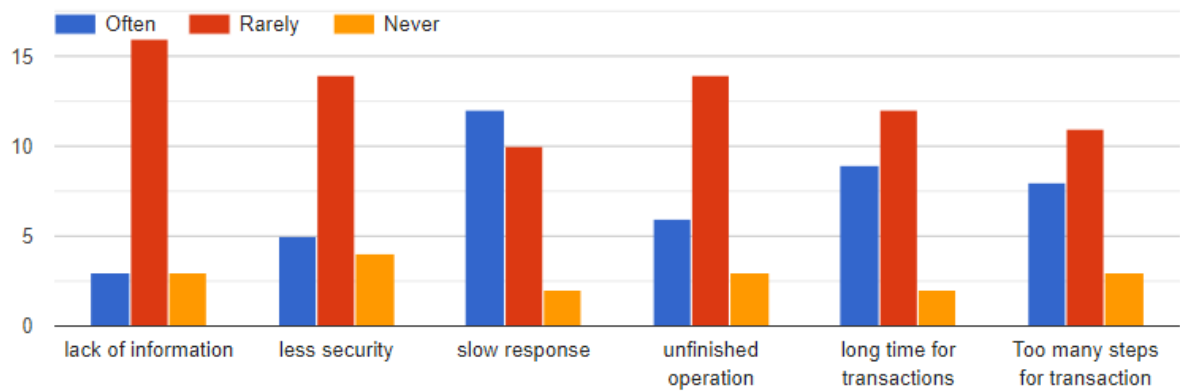
7. ATM Problems.

Ans.



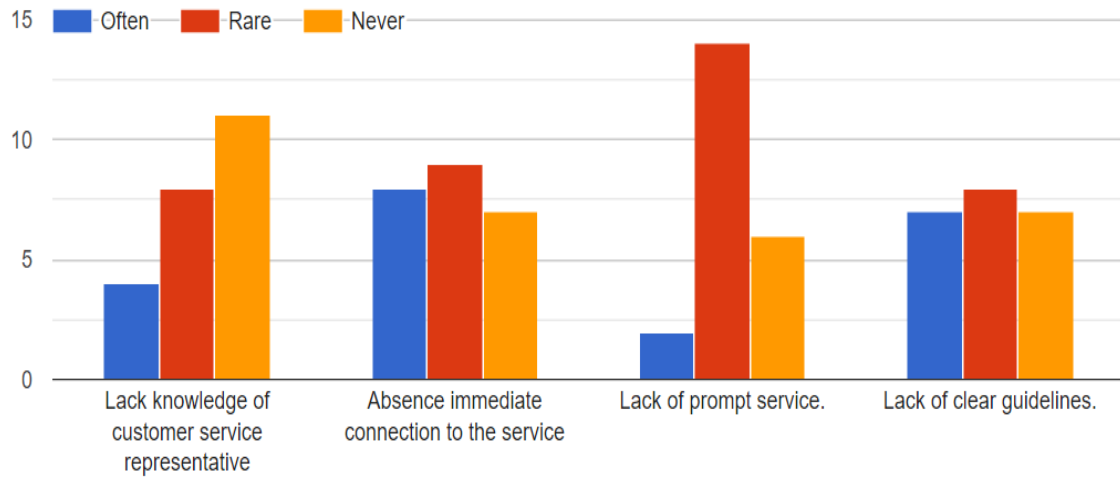
8. Internet Banking Problems.

Ans.



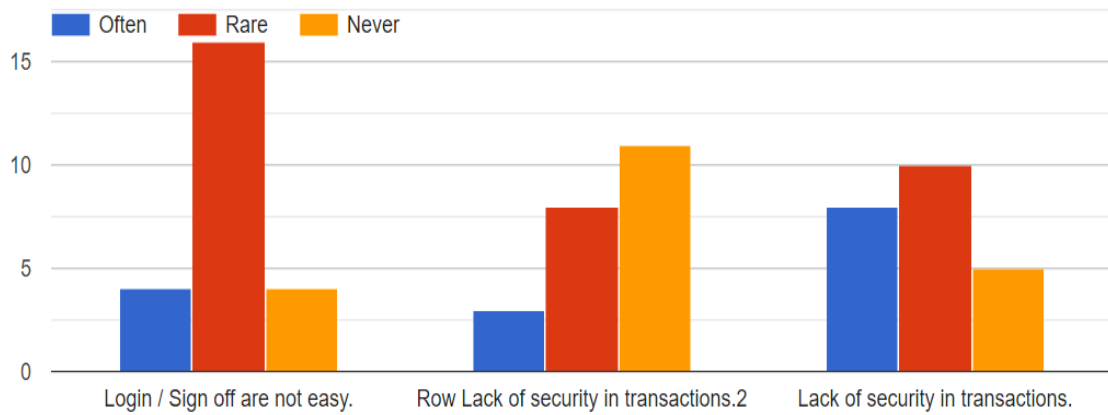
9. Telephone Banking Problems.

Ans.



10. Mobile Banking Problems.

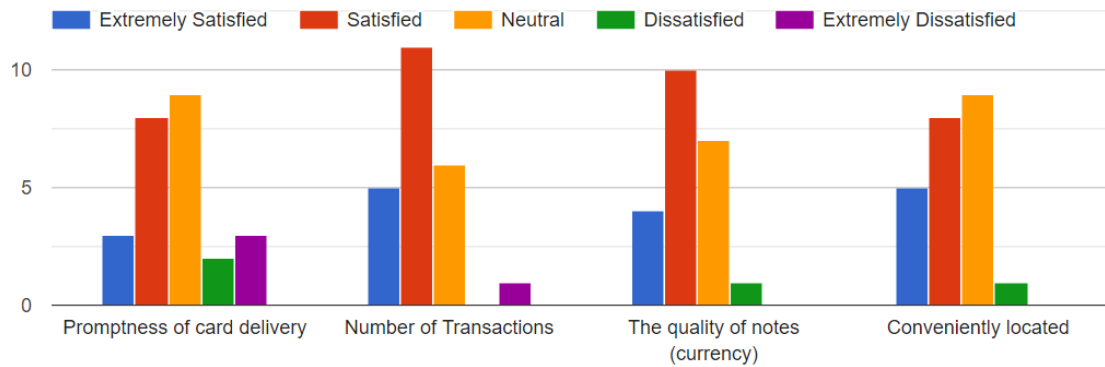
Ans.



Satisfaction on Technology usage

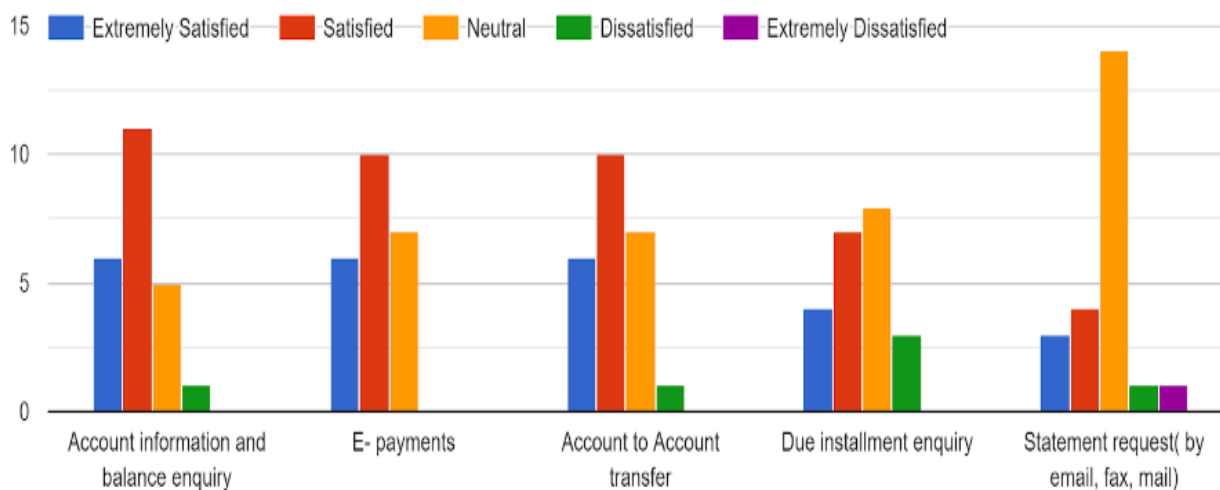
11. ATM Services

Ans.



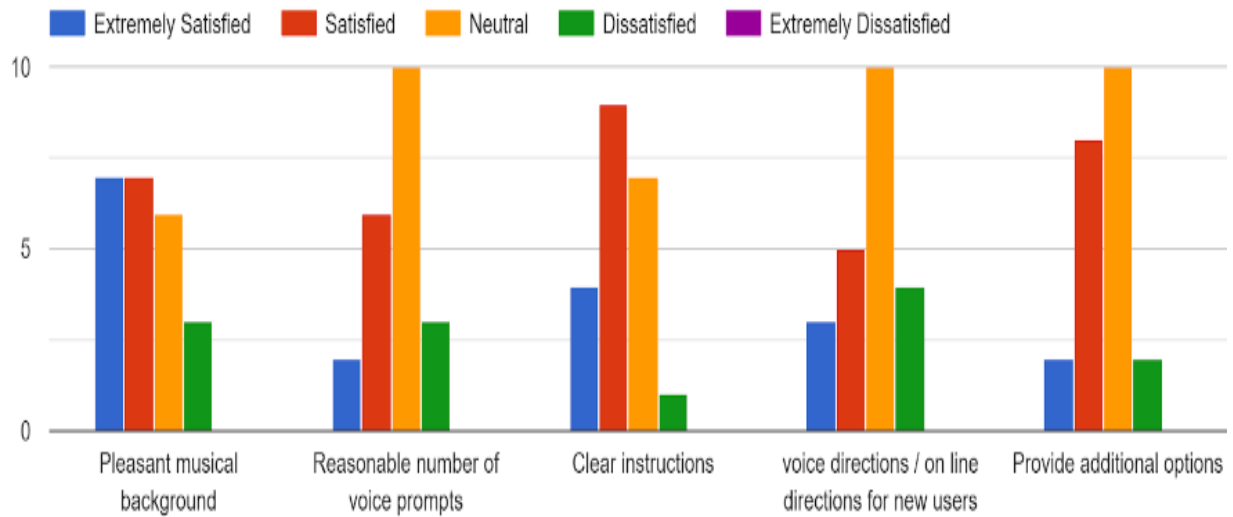
12. Internet banking Services

Ans.



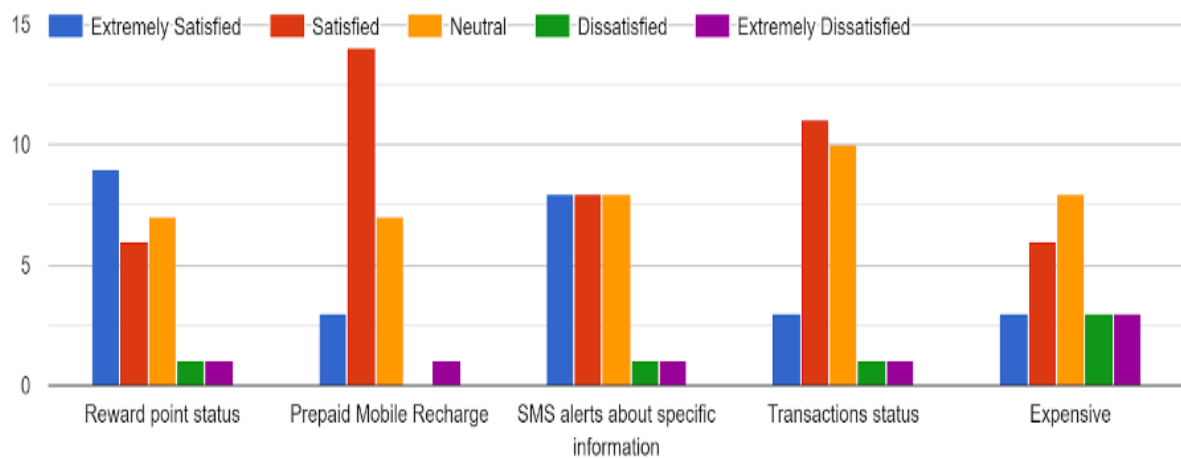
13. Telephone Banking Services.

Ans.



14. Mobile Banking services

Ans.



CHAPTER 6

CONCLUSION

Indian Banking systems have been with us for as long as people have been using money. Indian Bank and other financial institutions provide security for individuals, businesses and governments, alike. Let's recap what has been learned with this tutorial: In general, what banks do is pretty easy to figure out. For the average person banks accept deposits, make loans, provide a safe place for money and valuables, and act as payment agents between merchants and banks. Indian Bank are quite important to the economy and are involved in such economic activities as issuing money, settling payments, credit intermediation, maturity transformation and money creation in the form of fractional reserve banking. To make money, banks use deposits and whole sale deposits, share equity and fees and interest from debt, loans and consumer lending, such as credit cards and bank fees. In addition to fees and loans, banks are also involved in various other types of lending and operations including, buy/hold securities, non-interest income, insurance and leasing and payment treasury services.

History has proven banks to be vulnerable to many risks, however, including credit, liquidity, market, operating, interest rate and legal risks. Many global crises have been the result of such vulnerabilities and this has led to the strict regulation of state and national banks. However, other financial institutions exist that are not restricted by such regulations. Such institutions include: savings and loans, credit unions, investment and merchant banks, shadow banks, Islamic banks and industrial bank

BIBLIOGRAPHY

Reports

- Ahmad, K., “Bankers’ perception of electronic banking in Pakistan”, Journal of internet banking and commerce, April 2018
- Aladwani, A.M., “Online banking:A field study of drivers, development challenges And expectations”, International Journal of information Management .
- Agboola A. A ., “Electronic payment systems and Tele banking Services in Nigeria”, Journal of Internet Banking and commerce
- Eyadat, M. and Kozak, S., “The role of Information Technology in the profit and cost efficiency improvements of the banking sector”, Journal of Academy of Business and Economics,
- M.G, & Gebba T.R (2020) banking adoption an examination of TAM and Theory of Behavior, International Journal of Business Research and Development.
- Cano M.D & Domenech-Asensi,G (2021). A Security energy- efficient m-banking application for mobile devices. The journal software.

Internet sources

<https://www.wikipedia.org/>

<https://www.bankingfinance.in/impact-of-information-technology-in-indian-banking-industry.html>

<https://www.rbi.org.in>

<http://www.banknetindia.com/>

<https://www.bankingfinance.in>

https://www.hugedomains.com/domain_profile.cfm?d=india-bank&e=com

<https://www.capgemini.com/resources/top-ten-trends-in-banking-2017/>

QUESTIONNAIRE

SECTION 1: Demographic Details

1. Name (Optional): _____

2. Age:

- ☐ Below 20
- ☐ 20-30
- ☐ 31-40
- ☐ 41-50
- ☐ Above 50

3. Gender:

- ☐ Male
- ☐ Female
- ☐ Other

4. Occupation:

- ☐ Student
- ☐ Employee (Private Sector)
- ☐ Employee (Public Sector)
- ☐ Business Owner
- ☐ Retired
- ☐ Other: _____

5. Type of Bank You Prefer:

- ☐ Public Sector Bank
- ☐ Private Sector Bank

SECTION 2: Technology in Banking

6. **Which category of banks do you consider the most technologically advanced?**

- ☐ Public Sector Banks
- ☐ Private Sector Banks

7. **Which attribute of the bank do you value the most?**

- ☐ Trustworthiness and reliability
- ☐ Quality of customer service
- ☐ Technological advancements
- ☐ Accessibility and convenience

8. **Which factor encourages you to use new banking technologies?**

- ☐ Ease of transaction
- ☐ Faster service
- ☐ Security and safety
- ☐ Availability of multiple digital services

9. How familiar are you with the computer usage level in your bank?

- ☐ Very Familiar
- ☐ Somewhat Familiar
- ☐ Neutral
- ☐ Not Very Familiar
- ☐ Not Familiar at All

10. Which banking technology do you use the most?

- ☐ ATM Services
- ☐ Internet Banking
- ☐ Mobile Banking
- ☐ UPI & E-Payments
- ☐ Telephone Banking

11. How frequently do you use the following banking services per month?

Service	Daily Weekly Monthly Rarely Never				
ATM Services	☐	☐	☐	☐	☐
Internet Banking	☐	☐	☐	☐	☐
Mobile Banking	☐	☐	☐	☐	☐
Telephone Banking	☐	☐	☐	☐	☐

SECTION 3: Problems Faced in Technology Usage

12. Have you faced any issues while using the following banking technologies?

Type of Issue	Yes	No
ATM Service Problems (Card not working, Cash not dispensed, etc.)	☐	☐
Internet Banking Issues (Login problems, transaction failure, etc.)	☐	☐
Telephone Banking Problems (Long response time, unclear instructions, etc.)	☐	☐
Mobile Banking Problems (App crashes, payment failures, etc.)	☐	☐

13. If yes, please describe the issues:

SECTION 4: Satisfaction with Technology Usage

14. How satisfied are you with the following banking services?

Service	Very Satisfied	Satisfied	Neutral	Dissatisfied	Very Dissatisfied
ATM Services	☐	☐	☐	☐	☐
Internet Banking	☐	☐	☐	☐	☐
Mobile Banking	☐	☐	☐	☐	☐
Telephone Banking	☐	☐	☐	☐	☐

15.Do you think banks should improve their technological services?

- ☐ Yes
- ☐ No

16.What additional features would you like to see in banking technology?